

ALGORITHMIC SOCIAL INJUSTICES: ANTECEDENTS AND MITIGATIONS¹

Hüseyin Tanriverdi and John-Patrick Olatunji Akinyemi

McCombs School of Business, University of Texas at Austin
Austin, TX, U.S.A. {huseyin.tanriverdi@mcombs.utexas.edu} {johnpatrick.akinyemi@mcombs.utexas.edu}

A key assumption in data science is that the fairness of an algorithm depends on its accuracy. Antecedents that create accuracy problems are expected to reduce fairness and cause algorithmic social injustices. We theorize why complexities in ground truths, IT ecosystems, and statistical models of algorithms can also generate algorithmic social injustices, above and beyond the indirect effects of antecedents, through the mediation of accuracy problems. We also theorize technology design and organizational mitigation mechanisms for taming such complexities and reducing algorithmic social injustices. We tested the proposed theory in a sample of 363 matched pairs of problematic and problem-free algorithms. We found that complexities in ground truths affected algorithmic social injustices directly rather than through the mediation of accuracy problems. Failures in complex IT ecosystems of algorithms did not affect the likelihood of algorithmic social injustices, but they caused damage directly and indirectly through the mediation of accuracy problems. Failures in complex statistical models significantly increased algorithmic social injustices both directly and indirectly through the mediation of accuracy problems. The results indicate that agentic algorithms produce social injustices not only through accuracy problems but also through complexities in their ground truths, IT ecosystems, and statistical models. The proposed complexity taming mechanisms are effective in reducing algorithmic social injustice risks through (1) the user organization's quality in managing the algorithm's stakeholders, (2) designing algorithms with a large scope of human-like interaction capabilities, (3) the developer organization's algorithmic risk mitigations, and (4) the user organization's algorithmic risk mitigations.

Keywords: Machine learning, artificial intelligence, algorithm, complexity, social injustices, bias, accuracy, ground truth, multiple ground truths, mitigation, control, irreducible uncertainty

Introduction

Injustice anywhere is a threat to justice everywhere.
– Martin Luther King, Jr., 1964

Social injustice—defined as the condition where people have unequal access to valued resources, services, and positions in society (Kerbo, 2012)—has become increasingly prevalent in digital technologies. Artificial intelligence (AI) and machine learning (ML) algorithms promise to reduce social injustices (Purdy, 2020). However, some AI and ML algorithms have unintentionally learned, reproduced, and amplified them, aggravating social injustices (U.S. Congress, 2022; Eubanks,

2018). Understanding the root causes of social injustices in AI and ML algorithms is crucial for designing interventions that reduce both their likelihood and impact. Identifying effective mitigations for these injustices is essential not only for fostering trust in algorithms but also for achieving business goals and contributing to the broader societal objective of reducing social injustices. Thus, we study: (1) antecedents of algorithmic social injustices, and (2) organizational mitigations for algorithmic social injustices.

An algorithm is considered intelligent if it learns from patterns in big data and dynamically adjusts its behaviors and outcomes (Benbya et al., 2021). Algorithmic social injustice,

¹ Likoebe Maruping was the accepting senior editor for this paper. Nachiketa Sahoo served as the associate editor.

also referred to as “algorithmic bias” or “algorithmic unfairness,” is defined as an algorithm’s systematic and unfair treatment of certain individuals or groups (e.g., protected classes) in favor of others when allocating access to valued resources, services, and opportunities in society (Friedman & Nissenbaum, 1996) without a justified reason for such unequal access (Kordzadeh & Ghasemaghaci, 2022).

A common assumption in data science is that accuracy problems reduce fairness and cause algorithmic social injustices (De-Arteaga et al., 2022). Accordingly, mitigations for algorithmic social injustices focus on statistical methods for addressing accuracy and fairness problems. However, as the literature review in Appendix Table A1 shows, the link between accuracy and fairness is not straightforward. Some studies suggest a trade-off, where higher accuracy comes at the expense of fairness (Gardner et al., 2022; Liang et al., 2022; Valdivia et al., 2021), while others indicate that both can be improved simultaneously (Friedler et al., 2019; Rodolfa et al., 2021).

Another assumption is that accuracy issues can mediate the fairness effects of other factors. For example, accuracy problems may arise from (1) issues in the ground truth (GT) data used for training, (2) glitches in the IT systems that collect data, or (3) models failing to represent complex decision-making processes adequately (De-Arteaga et al., 2022). The accuracy problems can, in turn, lead to algorithmic social injustices. However, the mediating role of accuracy in fairness outcomes has yet to be theorized and tested. The lack of a clear theoretical framework and empirical research on the relationships between antecedents, accuracy problems, and algorithmic social injustices limits organizations’ ability to design effective mitigations. We aim to address these gaps.

Intelligent algorithms are agentic systems (Baird & Maruping, 2021). They operate as part of complex adaptive sociotechnical ecosystems that present fundamental, irreducible uncertainty, which is distinct from reducible uncertainty because of the lack of information and irreducible statistical error due to randomness. Fundamental uncertainty is a property of complex systems whose smart agents interact in nonlinear ways to produce emergent, unpredictable behaviors (McDaniel et al., 2003). Randomness, i.e., the lack of pattern or predictability in events, can be quantified with probabilities and modeled statistically. In contrast, fundamental uncertainty cannot be quantified or modeled. Randomness is a “known unknown,” whereas fundamental uncertainty is an “unknown unknown” (Daniel & Daniel, 2018). Statistical mitigations designed to address known unknowns in fairness and accuracy problems are inadequate for addressing unknown unknowns arising from complexities

in algorithms’ ground truths, IT ecosystems, and statistical models. If unaddressed, unknown unknowns can create algorithmic social injustices. This motivates us to theorize organizational mitigations for taming complexities in GTs, IT ecosystems, and statistical models of intelligent algorithms.

In theorizing the antecedents, we build on complexity science to explain how and why complexities in an algorithm’s GT, IT ecosystem, and statistical model are likely to increase algorithmic social injustice risks. In theorizing the mitigations, we use three taming principles from complexity science (Clarke & Stewart, 1997). We explain how the following complexity taming mechanisms can reduce algorithmic social injustices: (1) improving the quality of relationships among the algorithm’s stakeholders, (2) aligning the goals of the stakeholders, and (3) increasing the stakeholders’ capacity to take joint action.

We tested our theory in a sample of 363 matched pairs of algorithms. Each pair contained one algorithm that experienced a big enough problem to be noticed and publicly discussed by the algorithm’s stakeholders and a matching algorithm whose characteristics were similar to those of the problematic algorithm but whose stakeholders did not report any major problems.

The results on the antecedents indicate that intelligent algorithms produce social injustices directly through fundamental uncertainties arising from the complexities in their GTs, IT ecosystems, and statistical models. The indirect effects through the mediation of accuracy problems receive partial support for model failure only. Specifically, the lack of a well-established GT or the multiplicity of competing GTs can increase algorithmic social injustice risks directly by creating complexities and fundamental uncertainties for algorithms’ learning processes and decision tasks. Complexities in IT ecosystems of algorithms do not directly affect the likelihood of algorithmic social injustices. But they lead to IT failures, which cause damage both directly and through the mediation of accuracy problems. Failures in statistical models due to complexity increase the likelihood, scope, and impact of algorithmic social injustices both directly and through the mediation of accuracy problems.

As for results on the mitigations, designing an algorithm to have a large scope of human-like interaction capabilities (vision, speech, cognition, emotion, and other sensory capabilities) significantly reduces the likelihood, scope, and impact of algorithmic social injustices. Beyond such technology design, both user organizations’ quality in managing stakeholders of algorithms and user and developer organizations’ respective organizational mitigations for algorithm risks significantly reduce the likelihood and impact of algorithmic social injustices.

Background on Structural Inequalities and Fairness In AI

Crenshaw's intersectional approaches, Bourdieu's social capital theory, Giddens's structuration theory, and social stratification and class theories from Weber to Durkheim to Marx, all acknowledge an unequal distribution of resources between people (Morrison, 2006). There are structural inequalities along group characteristics such as gender, race, religion, socioeconomic status, etc. Throughout history, class inequalities have stemmed from economic differences in wealth (Engels & Wischnewetzky, 1845), with close associations between race and class (Du Bois, 1935). Some inequalities emerged in the era of colonialism when colonizers exploited the colonized region's resources and imposed their religion, economic and cultural practices, and languages onto the colonized people (Zajko, 2021). In the 18th century, Linnaeus (1758) developed a taxonomy classifying people into four races based on physical, ethnic, and geographical differences, a classification that has evolved into today's notion of race (Blumenbach, 1969). Given the deep tradition of historical and structural inequalities in society (Payne et al., 2019), today's technological social inequalities likely reflect the existing structural inequalities, at least in part (Kennedy et al., 2021; Martin, 2019).

Digital Technology and Social Inequality

People's unequal access to valued resources, services, and societal positions also manifests in the design and usage of digital technologies. Designers of digital technologies have been predominantly white males (Brown, 2014). Underrepresentation of other races and genders has led to underrepresentation of the excluded groups' values, needs, and preferences in digital technologies, thus reinforcing the existing inequalities. The digital divide literature has identified a connection between unequal access to digital devices and economic inequality, referring to it as the first-level digital divide (van Deursen & van Dijk, 2019), in contrast to the second-level digital divide, which is associated with a lack of technological competencies and difficulties using digital technologies (van Dijk, 2020). Further, datasets used by algorithms do not adequately represent members of protected classes who may have a limited presence in the digital world. While mainstream groups present data oceans, excluded and protected classes are "data deserts" for algorithms.

Fairness in AI

The foregoing review implies that people's lack of access to algorithms' ecosystems (e.g., digital technologies, data,

developers, interface, and interaction capabilities) can unfairly constrain or undermine some groups' meaning-making capacities. Moreover, gaps in the collective interpretive resources of some groups can give other groups an unfair advantage (Fricker, 2007). Interpreting algorithms' fairness or social justice requires an understanding of what it means for an algorithm to be fair (Friedler et al., 2021). There are three barriers to addressing this question. First, there are many definitions of fairness, and there is no consensus across and even within different ontological approaches (Mehrabi et al., 2021). Second, the many fairness metrics that data scientists use are incompatible (Hutchinson & Mitchell, 2019). Third, there are computational limits to developing fair algorithms in certain settings (Teodorescu et al., 2021). We organize the complex landscape of fairness notions under three disciplinary notions.

Statistical notions of fairness: The data science community uses statistical notions of fairness. In a review of over 140 papers, Mehrabi et al. (2021) found that each definition can be described as giving similar predictions to similar individuals, treating different groups equally, or obtaining the best properties of groups' and individuals' fairness notions. The community delineates bias and discrimination and demonstrates how each is a source of unfairness (Hutchinson & Mitchell, 2019). Yet there are situations where discrimination, defined as human prejudice based on sensitive attributes, is justified and explained by attributes specific to the context (Mehrabi et al., 2021). Fairness is addressed by evaluating the handling of protected attributes with statistical criteria: e.g., demographic parity, equalized odds and opportunity, and fairness through unawareness (Hutchinson & Mitchell, 2019; Teodorescu et al., 2021).

Procedural and distributive notions of fairness: Organizational scientists and economists focus on procedural and distributive notions of fairness. Procedural fairness refers to the consistency, neutrality, precision, revocability, and representativeness of the procedure used by an algorithm to produce its decisions (Marcinkowski et al., 2020; Morse et al., 2022). Distributive fairness refers to how algorithmic decisions optimize equal benefits for different stakeholders and distribute scarce resources equitably among them (Morse et al., 2022).

Stakeholder notions of fairness: In all notions of fairness reviewed, the subjective perceptions and judgments of an algorithm's stakeholders ultimately play a key role in the evaluation of whether an algorithm's decisions treat the stakeholders fairly or not (Morse et al., 2022). Thus, we adopt stakeholders' perceptions of algorithmic fairness. Building on stakeholder theory, we define the salient stakeholder of an algorithm as any group or individual who affects or is affected by the achievement of an algorithm's objectives (Mitchell et al., 1997).

Theoretical Foundations and Hypotheses ■

Intelligent algorithms are agentic information systems that have the capacity to learn, adapt, and act autonomously (Baird & Maruping, 2021). We build on complexity science in viewing agentic algorithms as complex adaptive systems (CAS). An agentic algorithm's CAS is made up of various connected and interdependent sociotechnical elements. Sensory information technologies, IT infrastructures, and IT applications collect big data on user behaviors. Statistical and computational models of algorithms process big data inputs. Users and other stakeholders generate big data inputs for algorithms with their dynamically changing behaviors. Algorithms learn from the big data inputs to dynamically update their learning and decision rules. The learning processes transfer the rights and responsibilities of human agents to algorithms or vice versa (Baird & Maruping, 2021). By learning human agency, algorithms become agentic IS that can make decisions autonomously, even in unstructured situations. The interactions among the agentic elements are reciprocal, nonlinear, and produce fundamental uncertainties that lead to unexpected, unpredictable outcomes (Anderson, 1999; Holland, 1995). Even if managers have perfect knowledge of the past states of the CAS, they cannot form subjective probabilities or relative frequencies for future states due to unpredictable behaviors and nonlinear interactions of the autonomous agents (Daniel & Daniel, 2018).

We build on these ideas to explain how complexities in GTs, IT ecosystems, and statistical models drive algorithmic social injustice risks. Risk is defined as the likelihood and impact of loss (Tanriverdi & Ruefli, 2004). Thus, we define *algorithmic social injustice risk* as the likelihood and impact of an algorithm's systematic and unfair treatment of certain individuals or groups (Friedman & Nissenbaum, 1996).

Ground Truth Status and Algorithmic Social Injustice Risks

Ground truth is the sum of all data collected, checked, and labeled in the context of a specific decision task for training an algorithm on how to make decisions on that task and calibrating its decision criteria, rules, and boundaries (C3.ai, 2022; Kozyrkov, 2021; Lexico, 2021).

Some algorithms have decision tasks with no well-established GTs or stakeholder groups and hold conflicting GTs about the decision tasks. Such algorithms face complexities and fundamental uncertainties in their learning processes and decision tasks. When there is a lack of a well-established GT, developers do not know or understand decision criteria, rules, or boundaries. Until the scientific process creates new knowledge to establish the GT for the decision tasks, developers

cannot effectively supervise the algorithm's learning process. They may have to resort to the use of unsupervised learning methods. Unlike supervised learning, unsupervised learning has relatively few fairness metrics. This limits developers' control over what the algorithm learns. If the algorithm learns biases and structural inequalities in the data, developers may not notice them until after victims notice and report the biases.

Alternatively, instead of using unsupervised learning, some developers attempt to reduce fundamental uncertainties in the decision tasks by asking the crowd to label the data and create the GT. However, in the absence of well-established decision criteria, rules, and boundaries, labelers resort to their own perceptions of truth and embed their implicit biases into GT labels (Babaei et al., 2021). The algorithm learns the resulting biases and treats some stakeholder groups systematically unfairly (Lopez, 2021).

Different labeler groups who hold multiple, irreconcilable GTs (Gordon et al., 2022) also interact in unpredictable ways during the GT elicitation process. For instance, one stakeholder group may label a social media post as hate speech, while another labels it as free speech (Cinelli et al., 2021; Kitchens et al., 2020). Most machine learning approaches view labeler disagreements as noise that must be eliminated rather than as useful signals. They attempt to eliminate the "noise" by a majority vote (Schaekermann et al., 2016). In a majority vote, the largest group's view overrides the minority groups' views and causes the algorithm to treat the excluded minority groups unfairly.

Some algorithmic decision tasks have well-established GTs (Smyth et al., 1994). By following established GT acquisition methods, developers can acquire the GT and reduce the uncertainty faced regarding the GT. However, mistakes can still happen. If GT labels are wrong, for instance, misclassification might be the correct action, and accuracy problems can emerge (Cooper et al., 2021). Accuracy problems, in turn, can lead to algorithmic social injustices (Gardner et al., 2022; Liang et al., 2022; Valdivia et al., 2021). Thus, accuracy problems can mediate the link between the GT status and algorithmic social injustices.

H1: *GT status of an algorithm affects algorithmic social injustice risks.*

H1a: *The lack of a well-established GT increases algorithmic social injustice risks.*

H1b: *Multiple conflicting GTs increase algorithmic social injustice risks.*

H1c: *Accuracy problems in an algorithm mediate the effect of the GT status on algorithmic social injustice risks.*

Failure in the Algorithm's IT Ecosystem and Algorithmic Social Injustice Risks

An IT ecosystem of an intelligent algorithm refers to IT infrastructures, IT applications, and IT services that collect, process, and store big data for the algorithm. Sensory information technologies (e.g., accelerometers, microphones, cameras, LIDAR) collect input data. Communication technologies transfer the sensed data to an algorithm. Storage, security, and privacy technologies enable the algorithm to manage the data. The connections, dependencies, and interactions among such IT elements make an algorithm's IT ecosystem complex. Complexity presents fundamental uncertainties and leads to the emergence of unexpected IT failures at the ecosystem level (Tanriverdi & Du, 2020). Some ecosystem-level IT failures affect marginalized populations disproportionately, wrongly depict them to algorithms, and amplify preexisting social injustices (Lopez, 2021).

For instance, Fitbit's fitness tracking algorithm initially relied on optical sensors to read vital signs from a person's skin and accelerometers to sense motion (Feehan et al., 2018). Optical sensors failed to read vital signs accurately from darker skin (Shcherbina et al., 2017). Thus, the IT failure disproportionately affected Black individuals. The victims expressed discontent about the inaccurate readings from their Fitbits and the subsequent injustices they experienced. Fitbit's accelerometer readings were initially also inaccurate for disabled or older people who walked with a walking aid or had a disease-related mobility limitation (Feehan et al., 2018). Due to the IT failures, the fitness algorithm made systematic errors in estimating the fitness levels of darker-colored races, disabled people, and the elderly, and amplified the preexisting social inequalities against them. Emergent IT failures are properties of complex IT ecosystems, and they present fundamental uncertainties for the accuracy and fairness performance of algorithms.

H2: *IT failure in an algorithm's IT ecosystem increases algorithmic social injustice risks.*

H2a: *Accuracy problems in an algorithm mediate the effect of an IT failure on algorithmic social injustice risks.*

Failure in the Algorithm's Statistical Model and Algorithmic Social Injustice Risks

We define a failure in an algorithm's statistical model as the model's inability to adequately represent the complex reality of a decision task. As the British statistician George Box

stated, "all models are wrong, but some are useful" (Box, 1979). All statistical models of algorithms are "wrong" in the sense that they are reductionist representations of the complex realities they attempt to represent. Some models are useful because their reductionist choices might adequately represent the complex reality.

The complexity of a decision task presents fundamental uncertainties for developers. For instance, developers may not know or understand which variables should be included or excluded in the reductionist model and how the intricate interactions might work between the included and excluded variables and affect outcomes. Such fundamental uncertainties can lead to discrepancies between the complex reality and its modeled version in the algorithm and create new social injustices.

For instance, in Idaho, Medicaid used an algorithm to automate health benefits allocation to individuals with a disability. The algorithm unfairly targeted 4,000 Idahoans and cut their benefits by as much as 30% (Stanley, 2017). A class-action lawsuit was filed against the state of Idaho's Medicaid. The court found that the algorithm produced arbitrary results for many people. In Arkansas, a similar algorithm unfairly treated Medicaid users. The algorithm's model took into account only a fraction of a lengthy list of items that human assessors used to rely on when making resource allocation decisions: "it turned out that a small number of variables could matter enormously for some people" (Lecher, 2018). With fewer support hours, individuals with cerebral palsy had to change their lifestyles. Some family members had to quit their jobs to care for loved ones with cerebral palsy due to the algorithmic injustices their family suffered (Lecher, 2018). These are examples of new social injustices that did not exist before the adoption of the algorithm.

A few studies found associations between model failures and AI bias (Anderson et al., 2019; Buolamwini & Gebru, 2018; Obermeyer et al., 2019), but they did not test if model failure had a direct effect on bias and/or an indirect effect through the mediation of accuracy problems. We hypothesize both direct and indirect effects to capture social injustices stemming, respectively, from the complexity of the modeling task and social injustices due to model accuracy problems.

H3: *A failure in an algorithm's model increases algorithmic social injustice risks.*

H3a: *Accuracy problems in an algorithm mediate the effect of model failure on algorithmic social injustice risks.*

Complexity Taming Principles

Complexity science offers three generic principles for taming complexity in systems: (1) improve the quality of relationships among the elements of the complex system, (2) align the goals of the elements, and (3) increase the capacity of the elements to take coordinated, system-level action (Clarke & Stewart, 1997). We build on these principles to theorize organizational mitigations for taming complexities in an agentic algorithm's sociotechnical ecosystem.

User Organization's Stakeholder Management Quality and Algorithmic Social Injustices

Algorithmic decisions potentially affect vast numbers of stakeholders who hold diverse sets of values, interests, and objectives. User organizations can potentially focus on the most salient stakeholder groups to tame complex dynamics among the stakeholders served. A salient stakeholder is defined as one with a power, legitimacy, or urgency relationship with an algorithm (Mitchell et al., 1997).

Salient stakeholders can have a variety of moral values and objectives, which may be difficult to reconcile or irreconcilable. To improve the quality of relationships among salient stakeholders, the user organization must proactively seek compromises. If an algorithm emphasizes the objectives and preferences of one group of salient stakeholders (e.g., men only, whites only, non-disabled users only, or right-handed people only), then algorithmic decisions could become less inclusive of the needs, preferences, and values of the other stakeholder groups (e.g., females, non-white race groups, disabled people, left-handed people, etc.). The excluded groups can perceive the algorithm's decisions to be unfair against themselves. User organizations can mitigate such algorithmic injustice risks by improving stakeholder relationships. We build on stakeholder theory to define the quality of stakeholder management as the extent to which a user organization of an algorithm: (1) creates value with and for an algorithm's salient stakeholders, (2) seeks to understand their social and political surroundings, (3) learns with and from them to create value opportunities, and (4) takes an integrative view of salient stakeholders by focusing on their interconnections (Freeman et al., 2017).

The quality of stakeholder management can satisfy the complexity of science principle of improving the quality of relationships among elements of a complex system. Discussing the diverse moral values, objectives, and decision criteria of the salient stakeholders and jointly seeking compromises can allow an algorithm to update its learning to remain inclusive of the salient stakeholders and make decisions that do not unfairly treat any of them.

H4: *User organization's stakeholder management quality reduces algorithmic social injustice risks.*

Algorithm's Scope of Human-Like Interaction Capabilities and Algorithmic Social Injustices

Algorithms often face a complex external environment comprising a broad spectrum of disabled and non-disabled users. Whether an algorithm's interaction capabilities are exclusive or inclusive of the broad spectrum of users could affect algorithmic social injustice risks. For instance, many chatbots provide only text-based interactions. They exclude disabled users who cannot use keyboards or touchscreens. According to complexity science, an organization needs to increase its internal complexity to match the external complexity (Page, 2010). Developers can accommodate a broad spectrum of disabled and non-disabled users in the external environment by endowing an algorithm with a broader scope of human-like interaction capabilities such as vision, speech, emotion, cognition, and other sensory capabilities such as touch, smell, and taste (Lake et al., 2017). In the chatbot example, endowing a chatbot with speech or vision capabilities can increase the chatbot's inclusivity of disabled users and reduce those users' perceptions of systematic and unfair treatment. For example, the famous physicist Stephen Hawking, who suffered from amyotrophic lateral sclerosis (ALS), gradually lost voluntary control of muscles and speech, severely limiting his ability to interact with computer systems. However, algorithms that offered touch, speech, vision, and eye-movement-based interaction capabilities were inclusive of him. He used a thumb switch and a blink switch attached to his glasses to control his computer. By squeezing his cheek muscles and "blinking," he was able to scan and select characters on the screen to compose speeches, surf the internet, send emails, and "speak" through a voice synthesizer (Access Computing, 2021). Designing a broad scope of human-like interaction capabilities can accommodate the needs of a broad spectrum of disabled and non-disabled users and reduce algorithmic social injustice risks (Costanza-Chock, 2020).

H5: *A large scope of human-like capabilities reduces algorithmic social injustice risks.*

Developer Organization's Governance and Controls and Algorithmic Social Injustice Risks

We define organizational governance and controls as a set of organizational structures, roles, policies, and procedures designed to reduce organizational risks (Tanriverdi & Du, 2020). In response to the proliferation of algorithmic problems such as social inequality, various organizational governance and control mechanisms have been proposed to mitigate algorithmic risks. Some examples include board-level oversight of algorithmic risks (Silverman, 2020); algorithmic transparency (Hosanagar & Jair, 2018); algorithmic accountability (U.S. Congress, 2022); algorithmic audits

(Guszcza et al., 2018; Morse et al., 2022); the use of FACT (fairness, accuracy, confidentiality, and transparency) principles by data science teams (RDS, 2022; van der Aalst et al., 2017); the diversity of data science team members' backgrounds, experiences, and perspectives (Moore et al., 2021); and governance and controls over the inputs, processing logic, and outputs of algorithms (Babel et al., 2019). However, their effectiveness has not been validated.

A developer organization of an algorithm is an organization that develops algorithms for internal use or sale to user organizations. Developer organizations often develop a portfolio of algorithms. For example, Google has developed a rich portfolio of algorithms, including advertising, search-matching, recommender, content moderation, photos, autonomous driving algorithms, etc. We frame the governance and controls of developer organizations at the level of the developer's overall portfolio of algorithms rather than any specific algorithm.

Governance and controls can potentially reduce social inequality risks by improving the quality of relationships among stakeholders, aligning stakeholders' goals, and building a capacity for stakeholders to take coordinated action in the design and use of algorithms. We illustrate the proposed nomological relationship by focusing on one of the proposed governance and control mechanisms, i.e., board-level oversight of algorithmic risks (Silverman, 2020).

Board-level oversight of algorithmic risks elevates the management of algorithmic risks to the highest organizational structure in the firm: i.e., the board of directors (Silverman, 2020). A board-level committee or a board member is charged with managing algorithmic risks. Many developers relegate algorithmic risk management to lower levels in the hierarchy, e.g., data science teams. This is problematic because an algorithm is a complex ecosystem with many stakeholders, including technologists, data scientists, users, regulators, the general public, etc. (Clarke & Stewart, 1997). Data scientists may not have full visibility into all stakeholders and their objectives, needs, preferences, and values. Moreover, a data science team may not know how the trade-offs among the conflicting objectives of powerful stakeholders ought to be made. The data science team may end up training an algorithm to optimize the objectives of some stakeholders at the expense of other stakeholders and thereby create social inequalities. Board-level oversight of algorithmic risks can potentially tame these challenges and reduce the likelihood of algorithmic social injustices because the board has better visibility of all stakeholders. The board also has decision rights to make trade-offs among the conflicting objectives of the stakeholders. The board is better positioned to improve the quality of relationships among the stakeholders, balance their conflicting interests,

achieve goal alignment, and foster coordinated action on algorithm design and use.

Similarly, other governance and controls can enable a developer organization to maintain high-quality relationships and goal alignment as the algorithm's ecosystem dynamically changes and evolves. Some developers announce governance and controls symbolically and do not follow through to implement the controls substantively. Substantive governance and controls are critical for reducing algorithmic social injustice risks.

H6: *Developer organization's substantive governance and controls reduce algorithmic social injustice risks.*

User Organization's Governance and Controls and Algorithmic Social Injustice Risks

Algorithmic social injustices often emerge in user organizations because developer organizations may not detect them during the development and test stages. A user organization is an organization that adopts an algorithm developed by a developer organization and implements it in its business operations to make decisions.

A user organization's governance and controls at the level of a specific algorithm can also affect whether algorithmic social injustices are likely to emerge or not during the usage phase of an algorithm. We focus on whether a user organization uses the following governance and control mechanisms: (1) Is the user organization committed to the ethical use of the algorithm? (2) Is the user organization transparent about how it uses the algorithm and how the algorithm makes decisions? (3) Does the user organization accept accountability for decisions made by the algorithm? Like the logic of the developer's governance and controls, the user's governance and controls can increase the quality of relations among stakeholders of the user organization's algorithm. They can align the stakeholders' goals and enable them to take coordinated action in response to unexpected, surprising challenges emerging during the usage phase of the algorithm.

H7: *The user organization's substantive governance and control mechanisms reduce algorithmic social injustice risks.*

Methods

Sample and Matching Criteria

We used a sample of problematic algorithms from the AI Algorithmic and Automation Incidents and Controversies (AIAAIC) repository, an independent public interest initiative

managed by a team of journalists and data scientists. For each problematic algorithm, the AIAAIC repository includes news articles or other sources discussing the alleged problem. Journalists validate the credibility of the sources. Data scientists validate whether the technologies discussed in the sources are indeed algorithms that have problems. The editorial team follows a governance process and applies four criteria (relevance, impact, credibility, and volume) to distinguish between “issues,” which are just potential concerns, and “incidents,” which are problems that already happened and caused negative impact. The AIAAIC repository has been used in peer-reviewed publications (Longoni et al., 2023), industry reports (Nestor et al., 2023), and EU policy reports (Molinari et al., 2021). We supplemented this repository by searching for additional problematic algorithms in Google, Factiva, EBSCOhost, Web of Science, and Google Scholar. We addressed the limitations of the AIAAIC repository by using conceptually grounded definitions for key concepts such as intelligent algorithm and algorithmic problem types (e.g., bias, IT failure, model failure, privacy breach, cybersecurity breach).

Inclusion Criteria

We used the following criteria to decide if a problematic algorithm in the AIAAIC repository qualified for inclusion in our study:

Algorithm: The problematic algorithm had to meet the definition of an intelligent algorithm (Benbya et al., 2021). It had to have the capability to learn from patterns in big data inputs and change its behavior over time based on changes in big data inputs.

Problem type: The problematic algorithm had to have one or more problems theorized in this study—i.e., bias, IT failure, or model failure. Although our study focused on nonmalicious operational IT failures, the repository also contained malicious IT failures, such as privacy and cybersecurity breaches in algorithms. We included malicious IT failures as controls.

Realized or potential problem: The problematic algorithm had to have a realized problem. We excluded entries that discussed potential algorithmic problems that had not yet emerged.

Usage status: When the problem emerged, the algorithm had to have been used with actual data and users during at least a pilot study or trial run, if not in full production. Entries that discussed problems detected during the ideation phases of algorithms were excluded.

Developer organization: The developer of the problematic algorithm had to be an organization. If an individual developed an algorithm, it was excluded.

User organization: The organization in which the algorithmic problem emerged had to be known. Entries that failed to specify or did not have a user organization were excluded.

Locations of user organizations: The user organization of the problematic algorithm had to be incorporated in the U.S. We excluded user organizations from international locations.

Matching Criteria

We formed a matched pair of problematic and problem-free algorithms. For each problematic algorithm, we searched for a problem-free competitor algorithm whose characteristics were very similar within the 12-month period before the emergence of the problematic algorithm’s problem. For instance, if Google Search was the problematic algorithm, Microsoft Bing was a competitor, and we checked to see whether it satisfied the following matching criteria:

Timing of match: The matching algorithm had to be in use as of the year of the problematic algorithm’s problem emergence. All matching criteria had to be satisfied as of that year.

Problem status: The matching algorithm had to be free of bias, IT failure, model failure, privacy breach, and cybersecurity breach problems. Stakeholders of the problem-free match did not discuss any of these problems within 12 months prior to the emergence of the problematic algorithm’s problem.

Function: The model of the problem-free algorithm had to be in the same functional category as the problematic algorithm (e.g., voice assistant algorithm, recommender algorithm, search-matching algorithm, etc.).

Application domain: The problem-free match had to be in the same application domain as the problematic algorithm (e.g., insurance, healthcare, HR, social media, transportation, etc.).

On-platform/Off-platform status: The problem-free match had to have the same on-platform/off-platform status as the problematic algorithm. On-platform algorithms had to work on multisided platforms (MSP); off-platform algorithms could not operate on an MSP.

For-profit/not-for-profit status: The problem-free match's user organization had to have the same for-profit/not-for-profit status as the problematic algorithm.

Public/private status: The user organization of the problem-free match had to have the same public/private sector status as the problematic algorithm.

Industry: The developer organization of the problem-free match had to have the same NAICS industry code as the developer of the problematic algorithm.

Fidelity of Match

76% of pairs had an "exact" match, as the two algorithms were the same in all matching criteria. The remaining 24% had a "coarse" match, as the two algorithms met all criteria except their application domains being slightly different. For instance, the problematic algorithm was a bar exam-proctoring algorithm; due to the lack of a competitor in the bar exam context, we selected a very similar proctoring algorithm in the college exam context.

The final sample had 363 pairs of problematic and problem-free algorithms, i.e., 726 algorithms from 18 industry segments and 150 functional categories in the U.S. between 2006 and 2022. Explanatory and control variables were coded within the 12-month period prior to the emergence of the algorithmic problem. Table 1 summarizes the sample construction process. For illustrative purposes, Appendix Table A2 shows three pairs of matched algorithms.

Source Documents

We relied on seven unique types of source documents: (1) technical documents and blogs prepared by developer organization's data science teams, (2) peer-reviewed academic publications, (3) investigative journalism articles (in which critical and in-depth journalist techniques are used to expose matters that are concealed behind circumstances that obscure understanding) (Kaplan, 2013), (4) court cases, (5) mainstream news articles, (6) developer organization documents and websites, (7) user organization documents and websites.

Table 1. Construction of a Matched Sample of Problematic and Problem-free Algorithms		
Step	Description of action taken	Size
1	Download problematic algorithms reported in the AIAAIC repository as of 07/21/2022*	878
2	Complement the AIAAIC sample with additional problematic algorithms found through keyword searches in Google, Factiva, EBSCOhost, and Web of Science	131
Subtotal of problematic algorithms before application of sample selection criteria		1009
3	Drop algorithms whose user organizations are not incorporated in the U.S.	611
4	Drop algorithms in the ideation phase that are not yet used with actual data and users	573
5	Drop algorithms failing to satisfy the definition of an intelligent algorithm (i.e., learn from patterns in big data inputs and change behaviors dynamically based on what it learns)	483
6	Drop algorithms that do not have any of the following problems: (1) bias, (2) IT failure, (3) model failure, (4) privacy breach, (5) cybersecurity breach	413
7	Drop algorithms that (1) were developed by an individual rather than an organization; (2) whose developer organizations were not specified	396
8	Drop problematic algorithms for which matching problem-free algorithms are not found	379
Subsample of problematic algorithms		379
9	For each problematic algorithm, go to the year of problem emergence and find a problem-free algorithm that satisfies (1) the sample selection criteria in steps 3-8; (2) has the same function, application domain, private/public status, for-profit/not-for-profit status, and on-platform/off-platform status as the problematic algorithm; and (3) has not developed: bias, IT failure, model failure, privacy breach, or cybersecurity breach.	379
Subsample of problem-free algorithms		379
Matched sample: Pairs of problematic (n1 = 379) and problem-free algorithms (n2 = 379)		758
10	Drop matched pairs for which problem-free or problematic algorithms have sources that lack credibility or insufficient sources to support claims.	726
Final matched sample: Pairs of problematic (n1 = 363) and problem-free algorithms (n2 = 363)		726

Note: *<https://www.aiaaic.org/aiaaic-repository/ai-and-algorithmic-incidents-and-controversies>

About 140 students contributed to collecting source documents needed for coding the study variables as part of class projects. However, they did not code the study variables. We reserved the coding tasks for two expert coders, as explained below. We gave students guidance for doing systematic keyword searches in the Factiva news database, company websites, annual reports, proxy statements, and Google searches to find source documents discussing characteristics of an algorithm and characteristics of developer and user organizations.

We considered peer-reviewed academic publications, investigative journalism, and court cases as credible since they publishers and by AIAAIC experts subject them through a rigorous process. For news articles, we took further measures to ensure the utmost quality. We used an independent media fact-checker (<https://mediabiasfactcheck.com/>), widely recognized by practitioners and academics, to analyze the credibility and factual reporting of the 192 unique news outlets in our sample. Two outlets had low factual reporting and credibility ratings, and a further three had only low credibility. A total of 12 news articles associated with these news outlets were dropped.

Most algorithms had between 11 and 16 unique sources across 27 data points for which source evidence was requested. Algorithms that did not have reliable or sufficient source documents were dropped from the sample. Table 2 reports the number of source documents used in coding the study variables. Appendix Table A3 lists the types of source documents collected to code user and developer organization characteristics (e.g., mitigations).

Coding Instrument Validation

We developed and validated a coding instrument for coding the study variables from the source documents. Definitions of variables were adapted from published literature or practitioner articles where no academic articles were available. Two Ph.D. students with degrees and professional experience in IS and strong training and research experience in data science served as independent expert coders who read the source documents to code the study variables by following the validated coding guidelines.

We established the reliability of the coding guidelines as follows. In each round of an iterative process, the two independent coders used the guidelines to code a small sample of algorithms (e.g., 20). We assessed the intercoder agreement rate after each episode of coding. After the first round, the agreement rate was 68%. After observing the two coders discuss the sources of coding discrepancies, it was clear that some variables were not tightly defined. We revised and tightened the definitions. After three rounds of iterations, the

intercoder agreement rate increased above the 90% threshold for establishing the reliability of the coding instrument. In the coding of 726 algorithms, agreement rates between the two independent coders were 95% for the dependent and 94% for the independent variables. The remaining discrepancies were jointly discussed and resolved by the two coders. To further assess the reliability of the coding guideline, we introduced two additional independent coders (a human-computer interactions graduate student and an undergraduate information systems honors student). They coded a small sample of 32 pairs and achieved a 96% intercoder agreement rate, reinforcing the reliability of the coding guideline instrument. Appendix Table A4 provides supporting literature and definitions of the study variables.

Dependent Variables

Risk is conceptualized in terms of likelihood and magnitude of loss (Tanriverdi & Ruefli, 2004). We used three dependent variables to measure the likelihood of algorithmic social injustices, the scope of social injustices, and the damage caused by algorithmic social injustices.

Algorithmic social injustices: Independent coders measured algorithmic social injustices with bias problems in an algorithm, i.e., by assessing whether the algorithm systematically and unfairly discriminated against certain individuals or groups of individuals in favor of others (Friedman & Nissenbaum, 1996). Federally protected classes were the target of algorithmic biases: i.e., age, sex (gender, pregnancy, sexual orientation, or gender identity), physical or mental disability, race, color, religion or creed, citizenship, national origin, or ancestry, and veteran status (Kappelman et al., 2019). As coders noted socioeconomic status as another target of algorithmic bias, we added it to the list. In source documents, coders looked for evidence that a stakeholder group claimed that the algorithm treated them systematically unfairly to reduce their access to valued resources, services, and positions. If such claims were made by one individual in isolation, it was not sufficient to conclude that the algorithm was biased. Coders triangulated the evidence by checking if two or more individuals within the same stakeholder group independently made similar allegations that the algorithm treated them systematically unfairly. After the coders measured if an algorithm had a given type of bias [1] or not [0], we took the logical OR of all types of algorithmic biases to measure whether the algorithm had a social injustice problem [1] or not [0].

Scope of social injustices: Some algorithms exhibited more than one type of algorithmic bias. We computed the total number of algorithmic biases to measure the scope of algorithmic social injustices.

Table 2. Source Documents Used to Code Study Variables

	Technical documents and developer blogs*	Peer-reviewed academic articles	Investigative journalism articles	Court cases	News articles	User org source pages	Dev org source pages	Total
Problematic algorithms	89	145	1214	44	972	794	2149	5407
Problem-free algorithms	92	120	649	13	825	501	2459	4659
Total URLs	181	265	1863	57	1797	1295	4608	10066

Note: *A total of 184 developer organizations made technical documents available. Some of them were in the form of peer-reviewed academic publications, and hence, they are reported in the next column.

Damage caused by algorithmic social injustices:

Independent coders used four items to measure whether the algorithmic social injustices harmed individuals or caused damage to the algorithm's user organization. The coders read the source documents covering the bias in an algorithm to code whether the bias harmed customers or employees of the user organization [1] or not [0]; whether harms to the customers or employees subsequently caused financial losses (e.g., regulatory fine, victim compensation) to the user organization [1] or not [0] or reputational loss (e.g., bad press, protests, or pressure on the organization to use accepted norms and meet standards) [1] or not [0]; and whether these harms led to lawsuits against the user organization [1] or not [0]. Cronbach's alpha was 0.85, indicating sufficient reliability.

Independent Variables

Ground truth status: Independent coders evaluated the status of ground truth used to train an algorithm and calibrate its decision-making rules and boundaries before the emergence of a problem in the algorithm: no well-established GT [0], well-established GT [1], or multiple conflicting GTs [2]. Then, we used [0] as the base case and created two dummy variables to represent well-established GT [1] and multiple conflicting GTs [1], respectively. Appendix Table A5 presents the evidence used to code the algorithm's ground truth status.

IT failure in the algorithm's ecosystem: Independent coders read the source documents to assess if there was any evidence of an operational IT glitch or malfunction in the algorithm's IT ecosystem prior to the emergence of the algorithmic bias [1] or not [0]. Appendix Table A5 presents the evidence used to code the IT failure variable.

Model failure in the algorithm: Data science literature attributes algorithmic biases to failures in statistical and computational models of algorithms (Mehrabi et al., 2021) (e.g., unrealistic or problematic assumptions, omission of important variables, variables that serve as proxies to protected classes, etc.). Independent coders read the source

documents to assess if there was any evidence of model failure in the algorithm [1] or not [0]. Appendix Table A5 illustrates evidence used to code the model failure variable.

Stakeholder management quality: Independent coders reviewed the source documents to assess if there was evidence that the user organization: created value with and for at least 50% of the algorithm's salient stakeholders [1] or not [0], sought to understand their social and political surroundings [1] or not [0], learned with and from them and used their criticism and feedback as value creation opportunities [1] or not [0], or took an integrative stakeholder view by focusing on the interconnections among the algorithm's salient stakeholders [1] or not [0]. Appendix Table A6 presents the evidence used to code the newly developed stakeholder management quality scale. Cronbach's Alpha, 0.69, is very close to the 0.70 threshold and thus satisfactory for a newly developed scale.

Algorithm's scope of human-like capabilities: The AI community seeks to build systems that learn and think like people (Lake et al., 2017). Specifically, the AI community aims to endow machines with five human-like capabilities: (1) vision, (2) speech, (3) emotion, (4) cognition, and (5) senses such as touch, taste, etc. (Nirenburg, 2017). The independent coders first used categorical variables to measure whether each capability was present in each algorithm [1] or not [0]. Then, we computed the total number of capabilities to measure the algorithm's scope of human-like capabilities. Appendix Table A7 presents the evidence used to code the scope of the algorithm's human-like capabilities.

Developer organizations' governance and controls: The independent coders reviewed the source documents of the algorithm's developer organization to code if there was evidence indicating that the developer organization had governance and controls to mitigate risks of its portfolio of algorithms. For each mitigation, the following scale was used: no evidence of the mitigation [0], symbolic evidence of the mitigation [1], and substantive evidence of the mitigation [2]. This measurement was taken for a total of eight developer organization mitigations: (1) board-level oversight of

algorithmic risks (Silverman, 2020), (2) commitment to ethical development and the use of AI (Leslie, 2019), (3) algorithmic transparency (Hosanagar & Jair, 2018), (4) algorithmic accountability (U.S. Congress, 2022), (5) algorithmic audits (Guszcza et al., 2018; Morse et al., 2022), (6) use of FACT (fairness, accuracy, confidentiality, and transparency) principles by data science teams (RDS, 2022; van der Aalst et al., 2017), (7) diversity of data science team members' backgrounds, and perspectives (Moore et al., 2021), and (8) governance and controls over inputs, processing logic, and outputs of algorithms (Babel et al., 2019). Cronbach's alpha for the eight items was 0.80, indicating sufficient reliability.

User organizations' governance and controls: In line with our theory, we measured the user organization's mitigations at the level of a specific algorithm. We used the same three-point measurement scale: no evidence of the mitigation [0], symbolic evidence of the mitigation [1], and substantive evidence of the mitigation [2]. Independent coders measured if the user organization had (1) committed to ethical use of the algorithm, (2) accepted accountability for the algorithm's decisions, and (3) exhibited transparency about the variables the algorithm used to make decisions. Cronbach's alpha for the three items was 0.71, indicating sufficient reliability. Appendix Table A8 presents the evidence used to code organizations' governance and controls.

Control Variables

To account for alternative explanations and potential endogeneity concerns, we included organizational-level and algorithm-level controls: *privacy breach* (present [1] or not [0]); *cybersecurity breach* (present [1] or not [0]); *algorithm's variety of stakeholders* (count of the algorithm's unique stakeholder groups); *private/public status of the algorithm's user organization* (private [1], public [0]); *for-profit/nonprofit status of the algorithm's user organization* (for profit [1], nonprofit [0]); *on-platform/off-platform status of the algorithm* (on-platform [1], off-platform [0]).

Algorithm's decision support mode: We evaluated the algorithms in the sample on the level of human augmentation in the algorithm's decision-making based on insights from Teodorescu et al. (2021). In realizing fairness in algorithm outcomes, Teodorescu et al. (2021) emphasized the importance of augmentation. If the algorithm processes a very high volume of transactions and makes a very high percentage of decisions in a short amount of time (e.g., credit card swipe approvals) and flags only a small percentage of those transactions as exceptions to be reviewed by humans (e.g., suspicion of fraud, or customer dispute), we coded it as *automation* [0]. We used automation as the base case and coded three additional dummy variables. If humans audited

the algorithm's recommendations, but the machine made the final decisions, we coded it as *machine makes final decision* [1]. If humans reviewed the algorithm's recommendations and made the ultimate decisions for a very high percentage of the decisions, we coded it as *human makes final decision* [1]. Lastly, if the algorithm and humans delegated decision tasks to each other and both made non-negligible percentages of the decisions, we coded it as *human-machine collaborates* [1].

Industry similarity: If the developer and user organizations were in the same industry, we coded *industry similarity* as 1, otherwise 0.

Fairness goal: If the developer organization had set fairness as a goal for its algorithms, we coded *fairness goal* as 1, otherwise 0. Due to the multiplicity of fairness definitions, we only looked for whether the developer organization set fairness as a goal, regardless of the measures it might have used for fairness.

Accuracy problem measures the ratio of incorrect decisions to the total number of decisions recommended by the algorithm (Lebovitz et al., 2021; Morse et al., 2022). In the year of an algorithmic problem emergence, if the stakeholders of an algorithm were discussing any accuracy problems in the algorithm, we coded *accuracy problem* as 1, otherwise 0.

Fidelity of matching: If the problematic and problem-free algorithms matched exactly in all eight matching criteria, we coded *fidelity of matching* as 1. When the granularity of the functions or the application domains of the algorithms differed slightly but still matched, we classified it as "coarse matching" and coded *fidelity of matching* as 0.

Table 3 presents correlations and descriptive statistics. Table 4 shows that the measurement items of multi-item constructs loaded on their factors and had low loadings on other factors.

Estimation Models

The first dependent variable (DV-1), *social injustice*, is a dichotomous variable. Thus, we used a binary logistic model to estimate the likelihood of algorithmic social injustices. The second dependent variable (DV-2), *scope of social injustice*, and the third dependent variable (DV-3), *damage to the user organization*, are continuous variables. Thus, we used OLS regressions for them. One identification concern was organization-level unobserved heterogeneity. Some developer organizations (e.g., Google, Facebook, etc.) had multiple problematic algorithms in our sample. We clustered standard errors by unique developer organizations to account for potential unobserved heterogeneity.

Table 3. Correlations and Descriptive Statistics

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
1 Social injustice	1.000																							
2 Scope of social injustices	.954**	1.000																						
3 Damages	.564**	.541**	1.000																					
4 Well established GT=[1]; No well-established GT=[0]	-.207**	-.206**	-.110**	1.000																				
5 Multiple conflicting GTs=[1]; No well-established GT=[0]	.176**	.196**	.105**	-.423**	1.000																			
6 IT failure in algorithm's ecosystem ([1]=Yes; [0]=No)	.0036	.0034	.273**	.067	-.084*	1.000																		
7 Model failure in algorithm ([1]=Yes; [0]=No)	.424**	.419**	.483**	-.087*	.130**	.0003	1.000																	
8 Stakeholder management quality	-.079*	.0061	-.106**	.152**	.0035	.0002	.0040	1.000																
9 Algorithm's scope of human-like capabilities	-.173**	-.167**	-.025	.0018	-.132**	.180**	-.0062	-.0020	1.000															
10 Developer organizations' governance and controls	-.105**	-.093**	-.096**	.0051	.0014	-.0054	.0002	.144**	-.0020	1.000														
11 User organizations' governance and controls	-.159**	-.142**	-.189**	.078*	-.0053	-.0038	-.071*	.117**	.0030	.281**	1.000													
12 Privacy breach in algorithm ([1]=Yes; [0]=No)	.214**	.191**	.339**	-.146**	-.0039	-.0031	.0036	-.068*	.104**	-.0061	-.122**	1.000												
13 Cybersecurity breach in algorithm ([1]=Yes; [0]=No)	.0010	.0008	.206**	.0019	.0058	.098**	.0025	-.0063	.0062	-.0014	-.0052	.083*	1.000											
14 Algorithm's variety of stakeholders	.0029	.0044	.147**	.0044	.106**	.110**	.111**	.112**	.080*	.066*	.0034	-.113**	.0064	1.000										
15 Private/public/private status ([1]: private; [0]: public)	.0040	.0034	.0019	.108**	-.0043	.0038	.0029	.0003	-.0010	.111**	.119**	-.099**	.091*	.083*	1.000									
16 For profit / Not profit status ([1]: for profit; [0]: not-for-profit)	-.075*	.0068	-.0003	.136**	-.0034	.0062	-.0004	.0022	-.0004	.110**	.096**	-.146**	.0049	.160**	.843**	1.000								
17 On-platform/off-platform status ([1]: on-platform; [0]: off-platform)	.106**	.115**	.0029	-.110**	.175**	-.122**	.0036	.0034	-.303**	.086**	.066**	-.157**	.0047	.201**	.356**	.409**	1.000							
18 Machine makes final decision ([1]: Yes; [0]: No)	.0005	.0008	.0034	-.0070	.0027	-.0026	.0029	-.0035	-.0008	-.0029	.0011	.0018	.0038	-.0005	-.096**	-.076*	-.0013	1.000						
19 Human makes final decision ([1]: Yes; [0]: No)	.0059	.0054	.0006	-.109**	.0058	-.0047	-.0032	-.0028	-.0003	-.0005	-.115**	.118**	-.0036	-.097**	-.361**	-.349**	-.269**	-.173**	1.000					
20 Human-machine collaborates ([1]: Yes; [0]: No)	-.112**	-.107**	.0008	.0024	.0005	.170**	-.0045	-.0010	.273**	-.0024	.0030	-.088*	.0040	.175**	.0064	.082*	-.143**	-.127**	-.208**	1.000				
21 Developer & user orgs in same industry ([1]: Yes; [0]: No)	.0053	.0049	.097**	-.0003	.0048	.0027	.0067	.0008	-.0050	.0051	.139**	-.0057	.125**	.142**	.460**	.432**	-.0065	-.236**	.159**	-.100**	.0059	1.000		
22 Accuracy problem in algorithm ([1]: Yes; [0]: No)	.416**	.389**	.414**	-.150**	.0012	.0062	.313**	-.069*	-.0003	-.0055	-.105**	.310**	.087*	-.0053	-.108**	-.187**	-.151**	.0042	.159**	-.100**	-.0054	1.000		
23 Fairness of AI is a stated goal of developer org ([1]: Yes; [0]: No)	.0033	.0030	-.110**	-.0010	-.0037	-.103**	-.0063	.0007	-.0040	.145**	.140**	-.0001	-.100**	-.074*	-.073*	-.077*	-.0027	.0020	.133**	-.0062	-.0055	.0061	1.000	
24 Fidelity of matching ([1]: Exact matching; [0]: Coarse matching)	.0051	.0052	.0006	.0016	.0093*	.0023	.0002	.0004	.074*	.0025	.0002	-.0042	-.0023	.111**	.117**	.121**	.0063	-.148**	-.0065	.095*	.084*	-.0030	-.0010	1.000
Mean	.030	.040	.033	.054	.013	.008	.024	.025	.034	.075	.088	.014	.006	.524	.081	.075	.035	.010	.022	.013	.065	.021	.014	.076
SD	.046	.072	.037	.050	.034	.027	.043	.030	.016	.044	.061	.035	.024	1.19	.039	.043	.048	.029	.042	.034	.048	.041	.035	.043
Min	0	0	0	0	0	0	0	0	0.2	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0
Max	1	5	1	1	1	1	1	1	1	2	2	1	1	9	1	1	1	1	1	1	1	1	1	1
N	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726	726

Note: Two-tailed test of significance; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Kendall's tau_b for variables that are measured on an ordinal scale.

Table 4. Factor Analysis and Reliability

Measurement items	Component				Reliability (Cronbach's alpha)
	1	2	3	4	
Developer org's FACT principles*	0.723	0.037	-0.019	0.151	0.80
Developer org's ethical AI development*	0.679	0.062	0.027	0.212	
Developer org's governance and controls*	0.658	-0.118	0.077	0.036	
Developer org's board oversight of AI risk*	0.649	0.003	0.122	0.039	
Developer org's algorithm audits*	0.645	-0.023	-0.044	-0.028	
Developer org's AI accountability*	0.645	-0.130	-0.014	0.254	
Developer org's diversity of AI teams*	0.557	-0.041	0.201	-0.043	
Developer org's AI transparency*	0.517	-0.106	0.092	0.246	0.85
Financial loss is caused by user org's biased AI**	-0.063	0.846	-0.076	0.039	
Harm is caused to people by user org's biased AI**	-0.023	0.845	-0.021	-0.208	
Reputational damage is caused by user org's biased AI**	-0.012	0.827	0.029	-0.212	
Lawsuit filed due to user org's biased AI**	-0.109	0.788	-0.078	0.040	0.69
User org communicates with AI stakeholders**	0.120	-0.068	0.758	0.039	
User org learns with and from AI stakeholders**	0.064	-0.120	0.734	0.117	
User org has integrative view of AI stakeholders**	0.063	0.035	0.700	0.008	
User org has relationships with AI stakeholders**	0.030	0.003	0.656	0.016	0.71
User org's AI transparency*	0.044	-0.114	0.117	0.773	
User org's AI accountability*	0.196	-0.021	0.030	0.760	
User org's ethical use of AI*	0.228	-0.119	0.019	0.737	
Extraction method: Principal component analysis.					
Rotation method: Varimax with Kaiser normalization.					

Note: *Measurement scale: No evidence [0], Symbolic evidence [1], Substantive evidence [2]. **Measurement scale: No [0], Yes [1]

Another concern was potential endogeneity. Developer and user organizations' mitigations could correlate with the model's error term. Both developer and user organizations may self-select into implementing mitigations due to an omitted variable, and their decisions may subsequently affect algorithmic social injustices. This means that both variables might be endogenous and can bias estimation results. We used the instrumental variable estimation technique to address the potential endogeneity problem. A valid instrumental variable must be correlated with the endogenous variables but not directly affect the dependent variable. We used the following instrument: the first proposal of the Algorithmic Accountability Act (AAA) in the U.S. in 2019. While this proposal was not signed into law, it could have affected developer and user organizations' adoption of mitigations by signaling that the government might hold companies accountable for algorithmic biases. The proposal was not likely to have had a direct effect on algorithmic social injustices. However, it could have had an indirect effect through its effects on developer and user organizations' mitigations. Thus, we classified all problematic algorithms into two groups based on the year the social injustice emerged. If the year of emergence was on or before (vs. after) 2019, the year of the AAA, the developer, and user organizations of an algorithm would have had relatively low (vs. high) awareness of algorithmic bias risks and would have been less (vs. more) likely to implement mitigations. Because of how we matched problem-free and problematic algorithms, we classified each problem-free algorithm in the same way

based on the year the social injustice emerged in the problematic algorithm. Results remained qualitatively the same in two-stage least squares (2SLS) analyses with this instrument variable.

Results

Table 5 presents the direct effects of the hypothesized antecedents and the controls. The control for accuracy problems in algorithms significantly increased the likelihood, scope, and damage of algorithmic social injustices. Above and beyond the effects of accuracy problems, the multiplicity of GTs, IT failure, and model failure problems had additional effects on the dependent variables. Table 6 presents Sobel tests (Baron & Kenny, 1986) that assessed the mediation effects of accuracy problems between the antecedents and algorithmic social injustices.

Relative to algorithms with no well-established GTs, algorithms with well-established GTs significantly reduced the likelihood of algorithmic social injustices. This relationship was not mediated by accuracy problems in algorithms. Similarly, algorithms with multiple conflicting GTs significantly increased the likelihood and scope of algorithmic social injustices. Accuracy problems in algorithms did not mediate these relationships. The results partially support H1.

IT failures in algorithms' IT ecosystems did not significantly affect the likelihood or scope of algorithmic social injustices, but they significantly increased the damage caused both directly and through the mediation of accuracy problems. These results partially support H2.

Model failures in algorithms had direct, significant positive effects on the likelihood, scope, and damage of algorithmic social injustices. They also had indirect effects on all three dependent variables through the mediation of accuracy problems. These results fully support H3.

As Table 5 shows, the user organization's stakeholder management quality significantly reduced the likelihood of algorithmic social injustices and the magnitude of damage caused. These results support H4. An algorithm's scope of human-like interaction capabilities significantly reduced the likelihood of algorithmic social injustices, the scope of social injustices, and the magnitude of damage caused to the user organization. These findings support H5. Both the developer and the user organizations' governance and control mechanisms over algorithmic risks significantly reduced the likelihood of algorithmic social injustices and the magnitude of damage they caused. These results support H6 and H7.

Table 5. Algorithmic Social Injustice, Scope of Social Injustice, and Damages: Antecedents and Mitigations

	DV-1: Social injustice				DV-2: Scope of social injustices			DV-3: Damages		
	Binary logistic				OLS			OLS		
	B	SE	Exp (B)	Sig.	B	SE	Sig.	B	SE	Sig.
(Constant)	-0.356	0.639	0.700	0.577	0.361	0.139	0.010	0.028	0.058	0.629
Well established GT=[1]; No well-established GT=[0]	-0.283	0.244	0.753	0.246	-0.056	0.055	0.312	0.032	0.023	0.167
Multiple conflicting GTs=[1]; No well-established GT=[0]	0.726	0.325	2.068	0.025*	0.334	0.080	0.001***	0.048	0.033	0.147
IT failure in algorithm's ecosystem ([1]=Yes; [0]: No)	0.136	0.394	1.146	0.730	0.059	0.090	0.513	0.355	0.037	0.001***
Model failure in algorithm ([1]=Yes; [0]: No)	1.855	0.240	6.389	0.000***	0.477	0.059	0.001***	0.342	0.025	0.001***
Stakeholder management quality	-0.750	0.388	0.472	0.053	-0.038	0.082	0.639	-0.125	0.034	0.001***
Algorithm's scope of human-like capabilities	-2.958	0.862	0.052	0.001***	-0.396	0.167	0.018*	-0.200	0.070	0.004**
Developer organizations' governance and controls	-0.615	0.271	0.540	0.023*	-0.086	0.058	0.138	-0.041	0.024	0.089+
User organizations' governance and controls	-0.389	0.195	0.678	0.047*	-0.054	0.043	0.205	-0.056	0.018	0.002**
Privacy breach in algorithm ([1]=Yes; [0]: No)	1.012	0.298	2.751	0.001***	0.186	0.072	0.010**	0.315	0.030	0.001***
Cybersecurity breach in algorithm ([1]=Yes; [0]: No)	-0.821	0.435	0.440	0.059+	-0.199	0.101	0.050*	0.150	0.042	0.001***
Algorithm's variety of stakeholders	0.048	0.100	1.049	0.630	0.020	0.022	0.356	0.040	0.009	0.001***
Private/public status ([1]: private; [0]: public)	0.095	0.503	1.099	0.851	0.082	0.115	0.477	-0.034	0.048	0.482
For profit / Not profit status ([1]: for profit; [0]: not-for-profit)	-0.323	0.454	0.724	0.477	-0.131	0.106	0.218	0.025	0.044	0.566
On-platform/off-platform status ([1]: on-platform; [0]: off-platform)	0.861	0.282	2.367	0.002**	0.180	0.062	0.004**	0.053	0.026	0.040*
Machine makes final decision ([1]: Yes; [0]: No)	-0.225	0.362	0.799	0.534	-0.037	0.084	0.660	0.029	0.035	0.409
Human makes final decision ([1]: Yes; [0]: No)	0.139	0.300	1.149	0.644	0.000	0.066	0.997	0.011	0.027	0.684
Human-machine collaborates ([1]: Yes; [0]: No)	-0.058	0.378	0.943	0.877	-0.033	0.079	0.674	0.058	0.033	0.074+
Developer & user orgs in same industry ([1]: Yes; [0]: No)	0.310	0.266	1.364	0.244	0.039	0.057	0.490	0.040	0.024	0.088
Accuracy problem in algorithm ([1]: Yes; [0]: No)	1.738	0.265	5.686	0.000***	0.358	0.066	0.001***	0.163	0.027	0.001***
Fairness of AI is a stated goal of developer org ([1]: Yes; [0]: No)	-0.169	0.342	0.844	0.621	0.018	0.070	0.797	-0.022	0.029	0.448
Fidelity of matching ([1]: Exact matching; [0]: Coarse matching)	-0.398	0.247	0.672	0.108	-0.101	0.056	0.071+	0.001	0.023	0.950
	-2 log likelihood			593.109	R		0.533	R		0.730
	Cox & Snell R square			0.330	R square		0.284	R Square		0.533
	Nagelkerke R square			0.469	Adj. R square		0.262	Adj. R square		0.519
					SE of estimate		0.258	SE of estimate		0.258

Note: Two-tailed test of significance; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 6. Sobel Test of Accuracy Problem's Mediation Between Independent and Criterion Variables of Interest

Panel I. Mediation between independent variables (IV) of interest and social injustice										
	IV's indirect effect on social injustice through mediation of accuracy problem						IV's direct effect on social injustice			Interpretation
Independent variable (IV)	A	S.E. (A)	B	S.E. (B)	Sobel	Sig.	C	S.E. (C)	Sig.	
GT_Well_Established	-0.56	0.25	-0.28	0.24	1.03	0.30 n.s.	-0.42	0.23	0.07 +	Direct effect only
GT_Multiple	-0.45	0.36	0.73	0.33	-1.09	0.28 n.s.	0.56	0.31	0.07 +	Direct effect only
ModelFailure	1.88	0.24	1.85	0.24	5.54	0.00 ***	2.25	0.23	0.00 ***	Both direct and indirect effects
PrivacyBreach	1.57	0.27	1.01	0.30	2.94	0.00 **	1.49	0.28	0.00 ***	Both direct and indirect effects
ITFailure	1.03	0.38	0.14	0.39	0.34	0.73 n.s.	0.38	0.39	0.33 n.s.	Neither direct nor indirect effect
CSecurityBreach	0.91	0.42	-0.82	0.43	-1.43	0.15 n.s.	-0.66	0.43	0.13 n.s.	Neither direct nor indirect effect
Panel II. Mediation between independent variables (IV) of interest and scope of social injustice										
	IV's indirect effect on scope of social injustice through mediation of accuracy problem						IV's direct effect on scope of social injustice			Interpretation
Independent variable (IV)	A	S.E. (A)	B	S.E. (B)	Sobel	Sig.	C	S.E. (C)	Sig.	
GT_Well_Established	-0.56	0.25	-0.06	0.05	1.06	0.29 n.s.	-0.08	0.06	0.14 n.s.	Neither direct nor indirect effect
GT_Multiple	-0.45	0.36	0.33	0.08	-1.20	0.23 n.s.	0.31	0.08	0.00 ***	Direct effect only
ModelFailure	1.88	0.24	0.48	0.06	5.64	0.00 ***	0.58	0.06	0.00 ***	Both direct and indirect effects
PrivacyBreach	1.57	0.27	0.19	0.07	2.46	0.01 *	0.29	0.07	0.00 ***	Both direct and indirect effects
ITFailure	1.03	0.38	0.06	0.09	0.65	0.52 n.s.	0.10	0.09	0.25 n.s.	Neither direct nor indirect effect
CSecurityBreach	0.91	0.42	-0.20	0.10	-1.48	0.14 n.s.	-0.16	0.10	0.13 n.s.	Neither direct nor indirect effect
Panel III. Mediation between independent variables (IV) of interest and damages										
	IV's indirect effect on damages through mediation of accuracy problem						IV's direct effect on damages			Interpretation
Independent variable (IV)	A	S.E. (A)	B	S.E. (B)	Sobel	Sig.	C	S.E. (C)	Sig.	
GT_Well_Established	-0.56	0.25	0.03	0.02	-1.25	0.21 n.s.	0.02	0.02	0.41 n.s.	Neither direct nor indirect effect
GT_Multiple	-0.45	0.36	0.05	0.03	-1.00	0.32 n.s.	0.04	0.03	0.26 n.s.	Neither direct nor indirect effect
ModelFailure	1.88	0.24	0.34	0.02	7.19	0.00 ***	0.39	0.02	0.00 ***	Both direct and indirect effects
PrivacyBreach	1.57	0.27	0.32	0.03	5.15	0.00 ***	0.36	0.03	0.00 ***	Both direct and indirect effects
ITFailure	1.03	0.38	0.36	0.04	2.57	0.01 **	0.38	0.04	0.00 ***	Both direct and indirect effects
CSecurityBreach	0.91	0.42	0.15	0.04	1.89	0.06 +	0.17	0.04	0.00 ***	Both direct and indirect effects

Note: A, B, C are respectively the beta coefficients of the paths between independent and mediator (accuracy problem) variables; mediator and criterion variables (social injustice / scope of social injustice / damages); and independent and criterion variables. SE(A), SE(B), and SE(C) are standard errors. Following covariates are included in all equations: i.e., Fairness, StakeholderManagementQuality, ScopeHumanLikeCapabilities, DeveloperMitigations, UserMitigations, NumberStakeholder, PrivateOrPublic, ForProfitOrNonProfit, OnPlatformOrOffPlatform, MachineMakesFinalDec, HumanMakesFinalDec, HumanMachineCollaboration, IndustrySimilarity, FidelityMatching. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; two-sided t -tests used for all Sobel tests

Discussion and Conclusion

This study generates new insights about the antecedent mechanisms through which algorithmic social injustices emerge. It also proposes and validates new technology design and organizational mitigation mechanisms for reducing algorithmic social injustices.

Contributions to Research on Antecedents of Algorithmic Social Injustices

The results regarding the antecedents support our thesis that fundamental uncertainties—stemming from the complexities of GT data, IT ecosystems, and statistical models—directly contribute to algorithmic social injustices. In contrast, we found only partial support for the prevailing view that accuracy issues mediate the relationships between these antecedents and algorithmic social injustices. These findings advance IS research by identifying fundamental uncertainties as a primary cause of algorithmic social injustices, suggesting a need for new theoretical frameworks that focus on mitigating fundamental uncertainties. Current data science mitigations primarily address reducible uncertainties. IS research should complement them with organizational mitigations aimed at the fundamental uncertainties arising from the complexities of intelligent algorithms.

Contrary to the prevailing assumption, accuracy problems do not mediate the links between GT status variables and algorithmic social injustices. Instead, the absence of a well-established GT and the existence of multiple competing GTs have direct effects on algorithmic social injustices. These results warn IS researchers that existing data science methods for GT elicitation are inadequate for reducing algorithmic social injustices stemming from the fundamental uncertainties inherent in GT data. Until scientific advancements produce a consensus GT for a given decision task, IS researchers should refrain from developing intelligent algorithms for such tasks. Confusing fundamental uncertainties with reducible uncertainties could lead to the use of GT elicitation processes and the creation of labeled datasets for algorithm training. However, the fundamental uncertainties in these datasets can still introduce implicit biases, irreconcilable conflicts among labelers, and, ultimately, biased decision-making.

While IT failures within an algorithm's ecosystem can lead to accuracy problems, these accuracy problems do not influence the likelihood or scope of algorithmic social injustices, nor do IT failures have a direct effect on such injustices. This challenges the prevailing view that IT failures are a primary cause of algorithmic social injustices. However, IT failures are

not irrelevant. Rather, when they occur, they exacerbate the damage caused by accuracy problems, both directly and indirectly. These results call for further IS research on the governance and controls of intelligent algorithms' IT ecosystems to ensure their reliable operation. A well-functioning IT ecosystem can mitigate the impacts of other antecedents that contribute to algorithmic social injustices.

Model failures, on the other hand, have both direct and indirect effects on algorithmic social injustices through the mediation of accuracy problems. This highlights that statistical models in intelligent algorithms contain both reducible and fundamental uncertainties. While data science methods can address reducible uncertainties, fundamental uncertainties remain largely unaddressed. These are the uncertainties that stem from the inherent complexity of the real world, which cannot always be captured in a model. Intelligent algorithms have agentic components endowed with rights and responsibilities to make autonomous decisions. However, the interactions among these autonomous components are not yet fully understood, particularly in terms of how they influence outcomes and each other. Developers face fundamental uncertainties when deciding which variables to include or exclude in statistical models. The discrepancy between the complex, multifaceted nature of reality and the simplified, reductionist models used by algorithms contributes to algorithmic social injustices, as critical factors may be overlooked or misrepresented. IS research must focus on developing organizational mitigations for taming these fundamental uncertainties and enabling algorithms to make fairer and more accurate decisions that better reflect the complexity of real-world issues.

Contributions to Research on Mitigations for Algorithmic Social Injustice Risks

This study makes both theoretical and empirical contributions to the literature on mitigation mechanisms for algorithmic social injustices. Most existing research on mitigations focuses on developing statistical techniques to address known unknowns in fairness and accuracy issues within algorithms (De-Arteaga et al., 2022). While the concept of unknown unknowns has also been acknowledged (Daniel & Daniel, 2018), statistical mitigations are insufficient to address these complexities. We began developing theory to address unknown unknowns arising from the complexities of GT data, IT ecosystems, and the statistical models underlying intelligent algorithms. While the fundamental uncertainties inherent in such complexities are assumed to be irreducible, our theory and results indicate that fundamental uncertainties can be tamed by enhancing the quality of interactions between

the algorithm and its stakeholders, between the user organization and the algorithm's key stakeholders, and within the organization's own mitigation strategies for managing algorithmic social injustice risks. Future IS research could build on our theory and further theorize organizational mitigations for taming fundamental uncertainties arising from the complexities of intelligent algorithms.

Addressing external complexity in agentic algorithms:

Agentic algorithms interact with diverse sets of users, including both disabled and non-disabled individuals. Algorithms that fail to account for the varied needs of these groups risk systematically excluding or unfairly treating protected classes, such as people with disabilities. Our theory and findings inform IS research that designing algorithms with a broad range of human-like interaction capabilities can improve inclusivity and reduce perceptions of unfair treatment. These design interventions tame the external complexity and fundamental uncertainties the algorithm faces and ensure fairer outcomes for protected classes.

Enhancing stakeholder management: Our theory and findings also inform IS research that effective management of relationships between a user organization and the algorithm's stakeholders also plays a critical role in reducing algorithmic social injustices. In the absence of a well-established GT, or when multiple conflicting GTs exist, a high-quality stakeholder management strategy allows for negotiation and compromise on the decision criteria, rules, and boundaries of the algorithm. Although algorithms may not fully align with every stakeholder's specific value, the transparency and involvement of stakeholders in the process reduce perceptions of unfair treatment. In our sample, organizations improved stakeholder relationship management by designing algorithms to create value for stakeholders, understanding their social and political contexts, and using stakeholder feedback as an opportunity for continuous learning.

Organizational mitigation mechanisms: Finally, our theory and findings inform IS research that, beyond the statistical mitigation mechanisms, the organizational mitigation mechanisms employed by both developer and user organizations proved effective in reducing algorithmic social injustices. Measurement properties of these new constructs indicate that the proposed governance and controls are co-varying manifestations of a coherent organizational-level risk mitigation strategy for taming complexity dynamics in an algorithm's complex sociotechnical system. Future IS research can build on our theory to further theorize organizational mitigations for algorithmic social injustices arising from fundamental uncertainties inherent in complex, intelligent, agentic algorithms.

Implications for Practice

Intelligent algorithms not only replicate existing structural inequalities in society; they also amplify these inequalities at scale and create new ones. A key implication of this study for data science practice is that these outcomes are not solely attributable to issues of accuracy arising from ground truth acquisition processes, or to failures in IT and statistical models. Instead, they stem from the inherent complexities of these antecedents. Statistical mitigations addressing the "known unknowns" in algorithmic accuracy and fairness are insufficient to address the "unknown unknowns" arising from these complexities. Organizations should not relegate these issues to data science teams. They must supplement statistical mitigations of data science teams with organizational mitigations that address the multifaceted challenges within the sociotechnical ecosystems of algorithms.

Developers should also recognize that automating a decision is viable only when a well-established GT exists. When no clear GT exists, or when multiple conflicting GTs are present, algorithms face fundamental uncertainty in their learning processes and decision-making tasks. In such cases, refraining from automation may be a prudent approach.

Furthermore, designing algorithms with a broad scope for human-like interaction increases their internal complexity. However, this also enables algorithms to adapt to external complexities by accommodating the diverse interaction needs of users, promoting inclusivity.

Boundary Conditions, Limitations, and Future Work

This study is subject to several boundary conditions and limitations. First, it focuses exclusively on biases as examples of social inequalities. Whether the theory and findings extend to other forms of social inequality remains an open question for future research. Second, the absence of a systematic database containing comprehensive data on a broad, cross-sectional sample of algorithms, along with information about the mitigation mechanisms employed by their developers and user organizations, poses a limitation. The availability of technical documentation is also constrained. To address these gaps, we conducted extensive manual data collection using publicly available source documents. However, further testing and validation of our theory will be needed as alternative data sources become available. Third, our study could not measure variables for all the years during which an algorithm existed. Longitudinal analyses, made possible by emerging datasets, could shed light on how time-varying algorithm characteristics influence the emergence of algorithmic social

injustices. Finally, this study focused only on the mediating role of accuracy in the relationships between antecedents and algorithmic social injustices. Our findings underscore a broader knowledge gap: the mechanisms beyond accuracy through which antecedents lead to algorithmic social injustices remain poorly understood. Addressing this gap represents a critical area for future research.

Acknowledgments

We thank the editorial team and members of the review team for providing constructive criticism and suggestions throughout the review process. We are grateful to Charlie Pownall, founding manager of the AIAAIC, for meeting with authors to clarify how AIAAIC journalists and data scientists distinguish AI issues and incidents. Sanif Prasla, Zhuoran Jiang, Hira Jamshed, and Nitya Prakash provided excellent research assistance for this project. We also thank students in Dr. Tanriverdi's undergraduate and graduate IT audit and security courses, whose course projects indirectly contributed to this project through the source documents they identified. We thank McCombs Research Excellence Funds for supporting the undergraduate research assistants of this research project.

References

- Access Computing. (2021). *Dr. Stephen Hawking: A case study on using technology to communicate with the world*. <https://www.washington.edu/accesscomputing/dr-stephen-hawking-case-study-using-technology-communicate-world>
- Agrafiotis, I., Nurse, J. R. C., Goldsmith, M., Creese, S., & Upton, D. (2018). A taxonomy of cyber-harms: Defining the impacts of cyber-attacks and understanding how they propagate. *Journal of Cybersecurity*, 4(1), Article ty006. <https://doi.org/10.1093/cybsec/tyy006>
- Alphabet. (2017). *Annual report (Form 10-K)*. <https://stocklight.com/stocks/us/information/nasdaq-goog/alphabet/annual-reports/nasdaq-goog-2017-10K-17569702.pdf>
- Alphabet. (2020). *Definitive proxy statement (Def 14a)*. https://www.sec.gov/Archives/edgar/data/1652044/000130817920000203/lgoog2020_def14a.htm
- Anderson, H., Boodhwani, A., & Baker, R. S. (2019). Assessing the fairness of graduation predictions. *EDM*.
- Anderson, P. (1999). Perspective: Complexity theory and organization science. *Organization science*, 10(3), 216-232. <https://doi.org/10.1287/orsc.10.3.216>
- ArgoAI. (2021). *Argo AI Safety Report*. <https://www.argo.ai/wp-content/uploads/2021/04/ArgoSafetyReport.pdf>
- Arthurs, N., & Alvero, A. (2020). Whose truth is the "ground truth"? College admissions essays and bias in word vector evaluation methods. In *Proceedings of the International Conference on Educational Data Mining*.
- Asatiani, A., Malo, P., Nagbøl, P. R., Penttinen, E., Rinta-Kahila, T., & Salovaara, A. (2021). Sociotechnical envelopment of artificial intelligence: An approach to organizational deployment of inscrutable artificial intelligence systems. *Journal of the Association for Information Systems*, 22(2), 325-252. <https://doi.org/10.17705/1jais.00664>
- Babaci, M., Kulshrestha, J., Chakraborty, A., Redmiles, E. M., Cha, M., & Gummadi, K. P. (2021). Analyzing biases in perception of truth in news stories and their implications for fact checking. *IEEE Transactions on Computational Social Systems*, 9(3), 3, 839-850. <https://doi.org/10.1109/TCSS.2021.3096038>
- Babel, B., Buehler, K., Pivonka, A., Richardson, B., & Waldron, D. (2019). *Derisking machine learning and artificial intelligence*. McKinsey. <https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/derisking-machine-learning-and-artificial-intelligence#/>
- Baer, T. (2019). *Understand, manage, and prevent algorithmic bias, a guide for business users and data scientists*. Apress.
- Baird, A., & Maruping, L. M. (2021). The next generation of research on IS use: A theoretical framework of delegation to and from agentic IS artifacts. *MIS Quarterly*, 45(1), 315-341. <https://doi.org/10.25300/misq/2021/15882>
- Baron, R. M., & Kenny, D. A. (1986). The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *Journal of Personality and Social Psychology*, 51(6), Article 1173. <https://doi.org/10.1037/0022-3514.51.6.1173>
- Belanger, F., Hiller, J. S., & Smith, W. J. (2002). Trustworthiness in electronic commerce: The role of privacy, security, and site attributes. *The Journal of Strategic Information Systems*, 11(3-4), 245-270. [https://doi.org/10.1016/S0963-8687\(02\)00018-5](https://doi.org/10.1016/S0963-8687(02)00018-5)
- Benbya, H., Pachidi, S., & Jarvenpaa, S. (2021). Special issue editorial: Artificial intelligence in organizations: implications for information systems research. *Journal of the Association for Information Systems*, 22(2), 281-303. <https://doi.org/10.17705/1jais.00662>
- Black, D. (2019). *IBM announces new Watson AI, IBM Cloud capabilities, customers* AI Wire. <https://www.aiwire.net/2019/10/21/new-watson-ai-ibm-cloud-capabilities-customers/>
- Blasch, E., Pham, T., Chong, C.-Y., Koch, W., Leung, H., Braines, D., & Abdelzaher, T. (2021). Machine learning/artificial intelligence for sensor data fusion—Opportunities and challenges. *IEEE Aerospace and Electronic Systems Magazine*, 36(7), 80-93. <https://doi.org/10.1109/MAES.2021.3053222>
- Blumenbach, J. F. (1969). *On the natural varieties of mankind: De generis humani varietate nativa*. Bergman Books.
- Box, G. E. P. (1979). Robustness in the strategy of scientific model building. In R. L. Launer & G. N. Wilkinson (Eds.), *Robustness in Statistics* (pp. 201-236). Academic Press. <https://doi.org/10.1016/B978-0-12-438150-6.50018-2>
- Brown, D. A. (2014). Diversity and the high-tech industry. *Alabama Civil Rights & Civil Liberties Law Review*, 6, 95-123
- Buolamwini, J., & Gebru, T. (2018). Gender shades: Intersectional accuracy disparities in commercial gender classification. In *Proceedings of the Conference on Fairness, Accountability and Transparency* (pp. 77-91).
- C3.ai. (2022). *Glossary: Ground truth*. <https://c3.ai/glossary/machine-learning/ground-truth/>
- Cerrato, P., Halamka, J., & Pencina, M. (2022). A proposal for developing a platform that evaluates algorithmic equity and accuracy. *BMJ Health & Care Informatics*, 29(1). <https://doi.org/10.1136/bmjhci-2021-100423>
- Cinelli, M., De Francisci Morales, G., Galeazzi, A., Quattrociochi, W., & Starnini, M. (2021). The echo chamber effect on social media. *PNAS*, 118(9), e2023301118. <https://doi.org/10.1073/pnas.2023301118>

- Cirillo, D., Catuara-Solarz, S., Morey, C., Guney, E., Subirats, L., Mellino, S., Gigante, A., Valencia, A., Rementeria, M. J., & Chadha, A. S. (2020). Sex and gender differences and biases in artificial intelligence for biomedicine and healthcare. *NPJ Digital Medicine*, 3(1), Article 81. <https://doi.org/10.1038/s41746-020-0288-5>
- City of Boston. (2021). *Acceptable use policy*. <https://www.boston.gov/departments/innovation-and-technology/acceptable-use-policy>
- Clarke, M., & Stewart, J. (1997). *Handling the wicked issues: A challenge for government*. University of Birmingham, Institute of Local Government Studies.
- Clarke, R. (1999). Internet privacy concerns confirm the case for intervention. *Communications of the ACM*, 42(2), 60-67. <https://doi.org/10.1145/293411.293475>
- Clearview. (2020). *Join our team*. Clearview.ai. <https://www.clearview.ai/careers>
- Clinton, L. (2020). *Cyber-risk oversight 2020*. National Association of Corporate Directors and the Internet Security Alliance.
- Colvonen, P. J., DeYoung, P. N., Bosompra, N.-O. A., & Owens, R. L. (2020). Limiting racial disparities and bias for wearable devices in health science research. *Sleep*, 43(10), Article zsaal59. <https://doi.org/10.1093/sleep/zsaal59>
- Cooper, A. F., Abrams, E., & Na, N. (2021). Emergent unfairness in algorithmic fairness-accuracy trade-off research. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (pp. 46-54). <https://doi.org/10.1145/3461702.3462519>
- Costanza-Chock, S. (2020). *Design justice: Community-led practices to build the worlds we need*. The MIT Press. <https://doi.org/10.7551/mitpress/12255.001.0001>
- Cutler, J. (2019). *San francisco DA taps AI to remove bias in charging*. Bloomberg Law. <https://news.bloomberglaw.com/us-law-week/san-francisco-da-taps-ai-to-remove-bias-in-charging>
- Daniel, P. A., & Daniel, C. (2018). Complexity, uncertainty and mental models: from a paradigm of regulation to a paradigm of emergence in project management. *International Journal of Project Management*, 36(1), 184-197. <https://doi.org/10.1016/j.ijproman.2017.07.004>
- Darvishi, H., Ciunzo, D., Eide, E. R., & Rossi, P. S. (2020). Sensor-fault detection, isolation and accommodation for digital twins via modular data-driven architecture. *IEEE Sensors Journal*, 21(4), 4827-4838. <https://doi.org/10.1109/JSEN2020.3033134>
- De-Arteaga, M., Feuerriegel, S., & Saar-Tsechansky, M. (2022). Algorithmic fairness in business analytics: directions for research and practice. *Production and Operations Management*, 31(10), 3749-3770. <https://doi.org/10.1111/poms.13839>
- Deloach, J., Caragea, D., & Ou, X. (2016). Android malware detection with weak ground truth data. In *Proceedings of the IEEE International Conference on Big Data* <https://doi.org/10.1109/bigdata.2016.7841008>
- Dignan, L. (2020). *Amazon halo band review: Creepy yet unobtrusive and useful for quantified self-health data junkies*. ZDNet. <https://www.zdnet.com/article/amazon-halo-review-fitness-tracker-band/>
- Du Bois, W. E. B. (1935). *Black reconstruction*. Russell & Russell.
- Duke, I. (2021). *Lee's famous recipe using artificial intelligence at drive-thru to combat pandemic induced issues*. Dayton Daily News. <https://www.daytondailynews.com/local/lees-famous-recipe-using-artificial-intelligence-at-drive-thru-to-combat-pandemic-induced-issues/>
- Eitel-Porter, R. (2020). Beyond the promise: Implementing ethical AI. *AI and Ethics*, 1(1), 73-80. <https://doi.org/10.1007/s43681-020-00011-6>
- Ellis, D. P., Whitman, B., Berenzweig, A., & Lawrence, S. (2002). The quest for ground truth in musical artist similarity. In *Proceedings of the 3rd International Conference on Music Information Retrieval*. <https://doi.org/10.7916/D80R9ZRG>
- Engels, F., & Wischniewetzky, K. (1845). *The condition of the working class in England in 1844*. Routledge.
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. Picador.
- Feehan, L. M., Geldman, J., Sayre, E. C., Park, C., Ezzat, A. M., Yoo, J. Y., Hamilton, C. B., & Li, L. C. (2018). Accuracy of Fitbit devices: Systematic review and narrative syntheses of quantitative data. *JMIR mHealth and uHealth*, 6(8). <https://doi.org/10.2196/10527>
- Ferrarin, G. (2021). *This Tiktok lawsuit is highlighting how AI is screwing over voice actors*. Vice. <https://www.vice.com/en/article/this-tiktok-lawsuit-is-highlighting-how-ai-is-screwing-over-voice-actors/>
- Freeman, R. E., Kujala, J., & Sachs, S. (2017). *Stakeholder engagement: Clinical research cases*. Springer. <https://doi.org/10.1007/978-3-319-62785-4>
- Fricker, M. (2007). *Epistemic injustice: Power and the ethics of knowing*. Oxford University Press. <https://doi.org/10.1093/analys/anp028>
- Friedler, S. A., Scheidegger, C., & Venkatasubramanian, S. (2021). The (im)possibility of fairness: different value systems require different mechanisms for fair decision making. *Communications of the ACM*, 64(4), 136-143. <https://doi.org/10.1145/3433949>
- Friedler, S. A., Scheidegger, C., Venkatasubramanian, S., Choudhary, S., Hamilton, E. P., & Roth, D. (2019). A comparative study of fairness-enhancing interventions in machine learning. In *Proceedings of the Conference on Fairness, Accountability and Transparency* (pp. 329-338). <https://doi.org/10.1145/3287560.3287589>
- Friedman, B., & Nissenbaum, H. (1996). Bias in computer systems. *ACM Transactions on Information Systems*, 14(3), 330-347. <https://doi.org/10.1145/230538.230561>
- Gardner, J. W., Gursoy, F., & Kakadiaris, I. A. (2022). Accuracy-fairness tradeoff in parole decision predictions: A preliminary analysis. In *Proceedings of the IEEE/ACM International Conference on Big Data Computing, Applications and Technologies*. <https://doi.org/10.1109/bdcat56447.2022.00047>
- Ghosh, S. (2017). *Youtube fixed an autocomplete bug that suggested "How to have s*x with your kids."* Business Insider. <https://www.businessinsider.com/youtube-fixed-an-autocomplete-bug-that-suggested-how-to-have-sx-with-your-kids-2017-11>
- GoFundMe. (2021). *Secure fundraising you can count on*. GoFundMe Trust & Safety. <https://www.gofundme.com/c/safety>
- Gordon, M. L., Lam, M. S., Park, J. S., Patel, K., Hancock, J., Hashimoto, T., & Bernstein, M. S. (2022). Jury learning: Integrating dissenting voices into machine learning models In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3491102.3502004>
- Guszcza, J., Rahwan, I., Bible, W., Cebrian, M., & Katyal, V. (2018). *Why we need to audit algorithms*. *Harvard Business Review*. <https://hbr.org/2018/11/why-we-need-to-audit-algorithms>
- Hankerson, D., Marshall, A. R., Booker, J., Elmimouni, H., Walker, I., & Rode, J. A. (2016). Does technology have race? In *Proceedings*

- of the CHI Conference Extended Abstracts on Human Factors in Computing Systems (pp. 473-486).
- Harris, A. (2017). *How crowdfunding platform GoFundMe has created a \$3 billion digital safety net*. Fast Company. <https://www.fastcompany.com/3067472/how-crowdfunding-platform-gofundme-has-created-a-3-billion-digital>
- Holland, J. H. (1995). *Hidden order: How adaptation builds complexity*. Addison-Wesley
- Hosanagar, K., & Jair, K. (2018). *We need transparency in algorithms, but too much can backfire*. Harvard Business Review. <https://hbr.org/2018/07/we-need-transparency-in-algorithms-but-too-much-can-backfire>
- Hutchinson, B., & Mitchell, M. (2019). 50 years of test (un) fairness: Lessons for machine learning. In *Proceedings of the Conference on Fairness, Accountability and Transparency*. <https://doi.org/10.1145/3287560.3287600>
- Instructure. (2021). *The study hall*. <https://www.instructure.com/resources?redirectSource=uf&redirectPath=/canvas/resources/all/hiring-for-diversity-in-engineering>
- Jiang, S., Kang, P., Song, X., Lo, B. P., & Shull, P. B. (2021). Emerging wearable interfaces and algorithms for hand gesture recognition: A survey. *IEEE Reviews in Biomedical Engineering*, 15, 85-102. <https://doi.org/10.1109/RBME.2021.3078190>
- Johnson, K. (2020). *Google researchers release audit framework to close AI accountability gap*. VentureBeat. <https://venturebeat.com/ai/google-researchers-release-audit-framework-to-close-ai-accountability-gap/>
- Kaplan, D. E. (2013). *Global investigative journalism: Strategies for support* (2nd ed.). The Center for International Media Assistance
- Kappelman, L. A., Walker, R. M., & Massey, H. J. (2019). *Expanding the definition of protected class*. Massachusetts Lawyers Weekly. <https://masslawyersweekly.com/>
- Kearns, M., Neel, S., Roth, A., & Wu, Z. S. (2019). An empirical study of rich subgroup fairness for machine learning. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 100-109). <https://doi.org/10.1145/3287560.3287592>
- Kennedy, H., Steedman, R., & Jones, R. (2021). Approaching public perceptions of datafication through the lens of inequality: A case study in public service media. *Information, Communication & Society*, 24(12), 1745-1761. <https://doi.org/10.1080/1369118x.2020.1736122>
- Kerbo, H. R. (2012). *Social stratification and inequality: Class conflict in historical, comparative, and global perspective*. McGraw-Hill.
- Kirchner, L. (2016). *Where traditional DNA testing fails, algorithms take over*. ProPublica. <https://www.propublica.org/article/where-traditional-dna-testing-fails-algorithms-take-over>
- Kitchens, B., Johnson, S. L., & Gray, P. (2020). Understanding echo chambers and filter bubbles: The impact of social media on diversification and partisan shifts in news consumption. *MIS Quarterly*, 44(4), 1619-1649. <https://doi.org/10.25300/MISQ/2020/16371>
- Koene, A., Clifton, C., Hatada, Y., Webb, H., Patel, M., & Machado, C. (2019). *A governance framework for algorithmic accountability and transparency*. Publications Office of the European Union. <https://doi.org/10.2861/59990>
- Kolodny, L. (2017). *Virtual nurse app Sense.Ly raises \$8 million from investors including the Mayo Clinic*. TechCrunch. <https://techcrunch.com/2017/02/14/virtual-nurse-app-sense-ly-raises-8-million-from-investors-including-the-mayo-clinic/>
- Kordzadeh, N., & Ghasemaghaei, M. (2022). Algorithmic bias: Review, synthesis, and future research directions. *European Journal of Information Systems*, 31(3), 388-409. <https://doi.org/10.1080/0960085X.2021.1927212>
- Kozyrkov, C. (2021). *MFML 043—Ground truth*. Retrieved on April 29, 2021, from https://www.linkedin.com/posts/kozyrkov_mfml-043-ground-truth-activity-6825095832530894848-eKoa/
- Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40. Article e253. <https://doi.org/10.1017/s0140525x16001837>
- Lebovitz, S., Levina, N., & Lifshitz-Assaf, H. (2021). Is AI ground truth really true? The dangers of training and evaluating AI tools based on experts' know-what. *MIS Quarterly*, 45(3), 1501-1526. <https://doi.org/10.25300/MISQ/2021/16564>
- Lecher, C. (2018). *What happens when an algorithm cuts your health care*. The Verge. <https://www.theverge.com/2018/3/21/17144260/healthcare-medicaid-algorithm-arkansas-cerebral-palsy>
- Leslie, D. (2019). *Understanding artificial intelligence ethics and safety: a guide for the responsible design and implementation of ai systems in the public sector*. SSRN. <https://doi.org/10.2139/ssrn.3403301>
- Lexico. (2021). *Ground truth*. Retrieved on April 22, 2021, from https://www.linkedin.com/posts/kozyrkov_mfml-043-ground-truth-activity-6825095832530894848-eKoa/
- Li, Y., De-Arteaga, M., & Saar-Tsechansky, M. (2022). When more data lead us astray: active data acquisition in the presence of label bias. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* (pp. 133-146). <https://doi.org/10.1609/hcomp.v10i1.21994>
- Liang, A., Lu, J., & Mu, X. (2022). Algorithmic design: Fairness versus accuracy. In *Proceedings of the 23rd ACM Conference on Economics and Computation*, pp. 58-59. <https://doi.org/10.1145/3490486.3538237>
- Linnaeus, C. (1758). *Systema naturae* (Vol. 1). Stockholm Holmiae.
- Longoni, C., Cian, L., & Kyung, E. J. (2023). Algorithmic Transference: People overgeneralize failures of AI in the government. *Journal of Marketing Research*, 60(1), 170-188. <https://doi.org/10.1177/00222437221110139>
- Lopez, P. (2021). Bias does not equal bias: A socio-technical typology of bias in data-based algorithmic systems. *Internet Policy Review*, 10(4). <https://doi.org/10.14763/2021.4.1598>
- Mack, H. (2017). *Sensely raises \$8m for AI-powered virtual nurse app, eyes large-scale partnerships*. MobileHealth News. <https://www.mobihealthnews.com/content/sensely-raises-8m-ai-powered-virtual-nurse-app-eyes-large-scale-partnerships>
- Marcinkowski, F., Kieslich, K., Starke, C., & Lünich, M. (2020). Implications of AI (un-)fairness in higher education admissions. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 122-130). <https://doi.org/10.1145/3351095.3372867>
- Martin, K. (2019). Ethical implications and accountability of algorithms. *Journal of Business Ethics*, 160(4), 835-850. <https://doi.org/10.1007/s10551-018-3921-3>
- McDaniel, R. R., Jr., Jordan, M. E., & Fleeman, B. F. (2003). Surprise, surprise, surprise! A complexity science view of the unexpected. *Health Care Management Review*, 28(3), 266-278. <https://doi.org/10.1097/00004010-200307000-00008>
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6). <https://doi.org/10.1145/3457607>

- Microsoft. (2020). *Fate: Fairness, accountability, transparency, and ethics in AI*. <https://www.microsoft.com/en-us/research/theme/fate/>
- Mitchell, R. K., Agle, B. R., & Wood, D. J. (1997). Toward a theory of stakeholder identification and salience: Defining the principle of who and what really counts. *Academy of Management Review*, 22(4), 853-886. <https://doi.org/10.2307/259247>
- Molinari, F., Van Noordt, C., Vaccari, L., Pignatelli, F., & Tangi, L. (2021). *AI Watch. Beyond Pilots: Sustainable Implementation of AI in Public Services*. Publications office of the European Union. <https://doi.org/10.2760/440212>
- Moore, J. H., Truong, V. Q., Robbins, A. B., Nicholson, D., & Williams-Devane, C. L. (2021). Ten important roles for academic leaders to promote equity, diversity, and inclusion in data science. *BioData Mining*, 14(1). <https://doi.org/10.1186/s13040-021-00256-9>
- Morrison, K. (2006). *Marx, Durkheim, Weber: Formations of modern social thought*. SAGE.
- Morse, L., Teodorescu, M. H. M., Awwad, Y., & Kane, G. C. (2022). Do the ends justify the means? Variation in the distributive and procedural fairness of machine learning algorithms. *Journal of Business Ethics*, 181(4). <https://doi.org/10.1007/s10551-021-04939-5>
- NED. (2020). *AI platform prevents workplace accidents*. New Equipment Digest.
- Nestor, M., Loredana, F., Brynjolfsson, E., Etchemendy, J., & Ligett, K. (2023). *The AI Index 2023 annual report*. Stanford University. <https://aiunplugged.io/wp-content/uploads/2023/09/Artificial-Intelligence-Index-Report-2023.pdf>
- Nirenburg, S. (2017). Cognitive systems: Toward human-level functionality. *AI Magazine*, 38(4). <https://doi.org/10.1609/aimag.v38i4.2760>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447-453. <https://doi.org/10.1126/science.aax2342>
- Padmanabhan, B., Sahoo, N., & Burton-Jones, A. (2022). Machine learning in information systems research. *MIS Quarterly*, 46(1), iii-xix. <https://doi.org/10.25300/MISQ/2022/16359>
- Page, S. (2010). *Diversity and complexity*. Princeton University Press.
- Payne, B. K., Vuletic, H. A., & Brown-Iannuzzi, J. L. (2019). Historical roots of implicit bias in slavery. In *PNAS*, 116(24). <https://doi.org/10.1073/pnas.1818816116>
- Presenti, J. (2021). *Facebook's five pillars of responsible AI*. Facebook. <https://ai.meta.com/blog/facebook-five-pillars-of-responsible-ai/>
- Proctorio. (2020). *Data security*. <https://proctorio.com/privacy/data-security>
- Purdy, M. (2020). Unlocking AI's potential for social good. *Harvard Business Review*. <https://hbr.org/2020/10/unlocking-ais-potential-for-social-good>
- Qoves. (2020). *Learn about facial aesthetics*. <https://www.qoves.com/>
- Rajabi, A., & Garibay, O. O. (2021). Towards fairness in AI: Addressing bias in data using GANS. In *Proceedings of the 23rd Human-Computer Interaction International Conference*.
- RDS (Responsible Data Science). (2022). *Fact (fairness, accuracy, confidentiality, and transparency)*. Retrieved on April 29, 2021, from <https://redasci.org/>
- Richardson, R., Schultz, J. M., & Crawford, K. (2019). Dirty data, bad predictions: how civil rights violations impact police data, predictive policing systems, and justice. *New York University Law Review Online*, May 2019, 193-233.
- Rocket Companies. (2019). *Our board of directors*. <https://www.rocketcompanies.com/our-team/board-of-directors/>
- Rodolfa, K. T., Lamba, H., & Ghani, R. (2021). Empirical observation of negligible fairness—Accuracy trade-offs in machine learning for public policy. *Nature Machine Intelligence*, 3(10), 896-904. <https://doi.org/10.1038/s42256-021-00396-x>
- Sales, B. (2020). *A Google search for "Jewish baby strollers" yields anti-Semitic Images. An extremist campaign may be to blame*. Jewish Telegraphic Agency. <https://www.jta.org/2020/09/25/united-states/a-google-search-leads-to-anti-semitic-images-it-may-be-an-extremist-campaign>
- Samonas, S., & Coss, D. (2014). The CIA strikes back: Redefining confidentiality, integrity and availability in security. *Journal of Information System Security*, 10(3). <https://doi.org/10.4236/jis.2014.54016>
- Sarlin, J. (2019). *Anti-vaccination conspiracy theories thrive on Amazon*. CNN. <https://edition.cnn.com/2019/02/27/tech/amazon-anti-vaccine-books-movies/index.html>
- ScaleAI. (2021). *How it works*. <https://scale.com/image#how-it-works>
- Schackermann, M., Law, E., Williams, A. C., & Callaghan, W. (2016). Resolvable vs. irresolvable ambiguity: A new hybrid framework for dealing with uncertain ground truth. In *Proceedings of the 1st Workshop on Human-Centered Machine Learning at SIGCHI*.
- Selbst, A. D., Boyd, D., Friedler, S. A., Venkatasubramanian, S., & Vertesi, J. (2019). Fairness and abstraction in sociotechnical systems. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. <https://doi.org/10.1145/3287560.3287598>
- Shcherbina, A., Mattsson, C. M., Waggott, D., Salisbury, H., Christle, J. W., Hastie, T., Wheeler, M. T., & Ashley, E. A. (2017). Accuracy in wrist-worn, sensor-based measurements of heart rate and energy expenditure in a diverse cohort. *Journal of Personalized Medicine*, 7(2). <https://doi.org/10.3390/jpm7020003>
- SEC (U.S. Securities and Exchange Commission). (2019). *Annual Report (Form 10-K)*. https://www.sec.gov/Archives/edgar/data/1564408/000156459020003323/snap-10k_20191231.htm
- Silverman, K. (2020). Why your board needs a plan for AI oversight. *MIT Sloan Management Review*. <https://sloanreview.mit.edu/article/why-your-board-needs-a-plan-for-ai-oversight/>
- Smyth, P., Fayyad, U., Burl, M., Perona, P., & Baldi, P. (1994). Inferring ground truth from subjective labelling of venus images. In *Proceedings of the International Conference on Neural Information Processing Systems*.
- Stanley, J. (2017). *Pitfalls of artificial intelligence decision making highlighted in Idaho ACLU case*. ACLU. <https://www.aclu.org/news/privacy-technology/pitfalls-artificial-intelligence-decisionmaking-highlighted-idaho-aclu-case>
- Tanriverdi, H., & Du, K. (2020). Corporate strategy changes and information technology control effectiveness in multibusiness firms. *MIS Quarterly*, 44(4), 1573-1617. <https://doi.org/10.25300/misq/2020/14223>
- Tanriverdi, H., & Ruefli, T. (2004). The role of information technology in risk/return relations of firms. *Journal of the Association for Information Systems*, 5(11). <https://doi.org/10.17705/1jais.00061>
- Teodorescu, M. H., Morse, L., Awwad, Y., & Kane, G. C. (2021). Failures of fairness in automation require a deeper understanding of human-ML augmentation. *MIS Quarterly*, 45(3). <https://doi.org/10.25300/MISQ/2021/16535>
- Teodorescu, M. H., Ordabayeva, N., Kokkodis, M., Unnam, A., & Aggarwal, V. (2022). *Determining systematic differences in*

- human graders for machine learning-based automated hiring (Working paper). Brookings Institution Center on Regulation and Markets.
- Tesla. (2018). *Autopilot support*. <https://www.tesla.com/support/autopilot>
- Thompson, C. (2016). *Here's how Tesla's autopilot works*. Business Insider. <https://www.businessinsider.com/how-teslas-autopilot-works-2016-7>
- Tinder. (2019). *Security at Tinder*. <https://policies.tinder.com/security/intl/en-gb/>
- Triche, J. H., & Walden, E. (2018). The use of impression management strategies to manage stock market reactions to IT failures. *Journal of the Association for Information Systems*, 19(4), 333-357. <https://doi.org/10.17705/1jais.00494>
- Twitch. (2021). *Transparency reports*. https://safety.twitch.tv/s/article/Transparency-Reports?language=en_US
- Uber. (2019). *2019 annual report*. https://s23.q4cdn.com/407969754/files/doc_financials/2019/ar/Uber-Technologies-Inc-2019-Annual-Report.pdf
- U.S. Congress. (2022). *H.R.6580—117th Congress (2021-2022): Algorithmic Accountability Act of 2022*. <https://www.congress.gov/bill/117th-congress/house-bill/6580/text>
- Valdivia, A., Sánchez-Monedero, J., & Casillas, J. (2021). How fair can we go in machine learning? Assessing the boundaries of accuracy and fairness. *International Journal of Intelligent Systems*, 36(4), 1619-1643. <https://doi.org/10.1002/int.22354>
- van der Aalst, W., Bichler, M., & Heinzl, A. (2017). Responsible data science. *Business & Information Systems Engineering*, 59(5), 311-313. <https://doi.org/10.1007/s12599-017-0487-z>
- van Deursen, A. J., & van Dijk, J. A. (2019). The first-level digital divide shifts from inequalities in physical access to inequalities in material access. *New Media & Society*, 21(2), 354-375. <https://doi.org/10.1177/1461444818797082>
- van Dijk, J. (2020). *The Digital Divide*. Polity Press.
- Vaswani, S., & Ahmed, M. O. (2021). *Learning from multiple annotators: A survey* (Working paper). https://vaswanis.github.io/AI_report.pdf
- Wallen, M. P., Gomersall, S. R., Keating, S. E., Wisløff, U., & Coombes, J. S. (2016). Accuracy of heart rate watches: Implications for weight management. *PLOS One*, 11(5). <https://doi.org/10.1371/journal.pone.0154420>
- Walmart. (2021). *2021 ESG annual summary report*. <https://corporate.walmart.com/content/dam/corporate/documents/purpose/environmental-social-and-governance-report-archive/walmart-2021-esg-annual-summary.pdf>
- Wang, T., Ke, H., Zheng, X., Wang, K., Sangaiah, A. K., & Liu, A. (2019). Big data cleaning based on mobile edge computing in industrial sensor-cloud. *IEEE Transactions on Industrial Informatics*, 16(2), 1321-1329. <https://doi.org/10.1109/TII.2019.2938861>
- Wiggers, K. (2020). *Uber researchers investigate whether AI can behave ethically*. VentureBeat. <https://venturebeat.com/ai/uber-researchers-investigate-whether-ai-can-behave-ethically/>
- Willingham, A. (2017). *How to know if a GoFundMe page is legit*. CNN. <https://www.cnn.com/2017/04/17/us/iyw-gofundme-accounts-legitimate-how-to-trnd/index.html>
- Yu, S., Chai, Y., Chen, H., Sherman, S., & Brown, R. (2022). Wearable sensor-based chronic condition severity assessment: An adversarial attention-based deep multisource multitask learning approach. *MIS Quarterly*, 46(3), 1355-1394. <https://doi.org/10.25300/misq/2022/15763>
- Yu, S., Chai, Y., Samtani, S., Liu, H., & Chen, H. (2023). Motion sensor-based fall prevention for senior care: A hidden Markov model with generative adversarial network approach. *Information Systems Research*, 35(1), 1-15. <https://doi.org/10.1287/isre.2023.1203>
- Zajko, M. (2021). Conservative AI and Social Inequality: Conceptualizing alternatives to bias through social theory. *AI & Society*, 36(3), 1047-1056. <https://doi.org/10.1007/s00146-021-01153-9>

About the Authors

Hüseyin Tanriverdi is an associate professor of information, risk, and operations management at the Red McCombs School of Business at the University of Texas at Austin. His research focuses on firm-level risk/return implications of IT strategies. On the return side, he studies digital business strategies for surviving and thriving in complex, hypercompetitive, and disruptive business ecosystems. On the risk side, he studies the IT-related risks of firms, such as cybersecurity and privacy risks, algorithmic risks, ethical dilemma risks, and how IT governance and control mechanisms could mitigate them. Hüseyin teaches courses on strategic IT management, IT governance for enterprise risk management and regulatory compliance, and management of emerging information technologies. His research has been published in information systems journals, such as *Information Systems Research*, *MIS Quarterly*, *Journal of the Association for Information Systems*, *Journal of Strategic Information Systems*, and *European Journal of Information Systems*, and in management journals, such as *Academy of Management Journal* and *Strategic Management Journal*. His publications have received Best Published Paper awards from the Organizational Communications and Information Systems Division of the Academy of Management and *Telemedicine Journal*. ORCID: 0000-0002-6958-4310

John-Patrick Olatunji Akinyemi is a Ph.D. candidate in information systems at the McCombs School of Business, University of Texas at Austin. His research focuses on understanding the sociotechnical dynamics emerging in digital platforms and online communities, with particular attention paid to AI bias, (climate) misinformation, and digital platform governance. John-Patrick has presented his work at prestigious conferences, including the Academy of Management Annual Meeting and the Hawaii International Conference on System Sciences. His accomplishments include the ICIS Paper-A-Thon Award (2024), Best Paper Track Award at HICSS (2023), and the Pre-ICIS Workshop of Information Security & Privacy 3rd Place Student Paper Award (2023). He has also participated in the AOM CTO Doctoral Consortium (2024) and the ICIS Doctoral Consortium (2024). His research has been supported by funding from the Sandia National Laboratories LDRD Grant. Also, as a Brumley Graduate Fellow at the Strauss Center for International Security and Law, he contributes to interdisciplinary discussions on the societal impacts of technology, further bridging his academic work with broader policy and societal considerations. ORCID: 0000-0003-4859-4904

Appendix

Table A1. Review of Literature on Antecedents of Algorithmic Social Injustices (Bias)		
Nomological Link	Theory/conceptual explanation of the nomological link	Empirical results on the nomological link
Antecedents → Accuracy problem → Bias	No theoretical explanation as to how and why accuracy might mediate the links between antecedents and bias.	No empirical test of accuracy's potential mediation role between antecedents and bias.
Accuracy problem → Bias	Conflicting conceptual arguments: some studies assume a trade-off between accuracy and fairness (bias) of AI (Gardner et al., 2022; Liang et al., 2022; Valdivia et al., 2021), while others challenge this assumption and argue that accuracy and fairness can be achieved simultaneously (Friedler et al., 2019; Rodolfa et al., 2021).	Mixed results. Some studies find a trade-off: e.g., optimizing accuracy, subject only to marginal fairness, leads to substantial bias (Kearns et al., 2019), while others fail to find a major trade-off (Rajabi & Garibay, 2021): e.g., large fairness gain is achieved with negligible accuracy cost (Kearns et al., 2019).
Ground truth → Accuracy problem	No theoretical explanation as to how and why the lack of a ground truth (GT) or the presence of multiple conflicting GTs affects AI's accuracy. The majority assume a GT exists (Smyth et al., 1994) and focus on how GT acquisition affects accuracy (Deloach et al., 2016; Ellis et al., 2002): e.g., if GT labels are wrong, misclassification might be the correct action and accuracy problem can emerge (Cooper et al., 2021).	No empirical test of how the lack of GT or multiple conflicting GTs might affect AI's accuracy. A case study reported the lack of a GT for a decision task (Lebovitz et al., 2021) but did not link it to AI's accuracy.
Ground truth → Bias	No theoretical explanation as to how and why a lack of GT or multiple conflicting GTs can lead to bias. The majority focus on how GT acquisition problems lead to bias (Deloach et al., 2016; Ellis et al., 2002; Vaswani & Ahmed, 2021): e.g., if labelers introduce bias into GT, AI becomes biased (Li et al., 2022; Teodorescu et al., 2022).	No empirical test of how the lack of GT or multiple conflicting GTs might affect AI bias. Lebovitz et al. (2021) recognized the lack of a GT, but did not test if it was the AI bias source. Arthurs and Alvero (2020) observed the existence of multiple GTs, but did not test if multiple conflicting GTs were the source of AI bias.
Model failure → Accuracy problem	No theoretical explanation as to how and why model failures might affect accuracy problems.	Mixed results. For instance, Buolamwini and Gebre (2018) found significant links between model failure and accuracy problems, whereas Obermeyer et al. (2019) did not.
Model failure → Bias	No theory to distinguish model failure and bias. The majority confound the two by assuming that bias is ingrained in model assumptions (De-Arteaga et al., 2022). Few who distinguish the two do not theorize how and why model failure might affect AI bias (Selbst et al., 2019).	A few empirical studies find associations between model failure and AI bias (Anderson et al., 2019; Obermeyer et al., 2019), but they do not test if model failure has a direct effect on bias and or an indirect effect through the mediation of accuracy problem.
IT failure → Accuracy problem	No theoretical explanation for the IT failure–AI accuracy link (Yu et al., 2022; Yu et al., 2023). Few recognize that corrupted data from sensors can degrade AI's functionality and accuracy (Darvishi et al., 2020).	Engineering research tests how sensor problems affect technical accuracy (Blasch et al., 2021; Wang et al., 2019). No social science empirical research yet. Practitioner examples suggest an association (Wallen et al., 2016).
IT failure → Bias	No theoretical explanation of how and why IT failures might affect AI bias. Some assert that sensor or other device failures in AI's ecosystem can lead to AI bias (Cerrato et al., 2022; Cirillo et al., 2020; Jiang et al., 2021), but they do not theorize how and why.	No empirical test yet. Practitioner examples suggest an association (Colvonen et al., 2020): e.g., pulse monitor sensors on Apple iWatch and accelerometer sensors on Fitbit led to bias against darker-skinned and elderly users (Feehan et al., 2018; Hankerson et al., 2016).

Table A2. Illustrative Examples of Matching Problematic and Problem-Free Algorithms

Algorithms in matched pair	Application domain	Function	Platform status	Profit status	Developer's NAICS Sector	Match timing
1a. Problematic: Uber dynamic pricing algorithm—racial discrimination problem in 2020	Transportation—Ride-sharing	Pricing algorithm	On-platform	For-profit	Information (51)	2020
1b. Problem-free: Hitch dynamic pricing algorithm	Transportation—Ride-sharing	Pricing algorithm	On-platform	For-profit	Information (51)	2020
2a. Problematic: COMPAS recidivism risk estimation algorithm—racial discrimination in 2013	Criminal justice	Risk assessment algorithm	Off-platform	Non-profit	Information (51)	2013
2b. Problem-free: Public safety assessment—criminal arrest and violent risk estimation algorithm	Criminal justice	Risk assessment algorithm	Off-platform	Non-profit	Information (51)	2013
3a. Problematic: NYPD facial recognition—racial discrimination in 2017	Policing	Facial recognition algorithm	Off-platform	Non-profit	P&T Services (54)	2017
3b. Problem-free: Austin PD facial recognition	Policing	Facial recognition algorithm	Off-platform	Non-profit	P&T Services (54)	2017

Table A3. Source Document Types for Coding User and Developer Organization Characteristics

Controls and methods document; quality statement; code of conduct page; community guidelines page; performance and accountability report; FACT sheet; stakeholder analysis documents; corporate governance guidelines; risk management documents; safety report; transparency report
Journal pages; support and FAQ; Response letters to gov't writings; internal news and announcements; case studies; pressroom pages; owner's manual; product and service description pages; consumer success stories
ESG reports; CSR reports; impact reports; sustainability reports
About us page; vision and mission statement page; who we are page; org description page
LinkedIn search results; career opportunities page; leadership and structure; "our culture" page; diversity & inclusion statement; DEI impact report; workforce diversity report; diversity committee charter
Investigations on the organization's use or development of algorithm; analysis of key organization documents; Glassdoor profile; interviews of prominent figures of the organization
Audit committee charter; shareholder meeting notes; committee compositions; board meeting agenda; oversight board structure; strategic plan
Privacy statements; compliance pages; ethics statements; security methods; data handling procedures; GDPR statement; data transparency policies; AI fairness and responsible practices; terms and conditions
10K annual statement; DS-1 statement; proxy statement (DEF14A); S-1; 20-F
Org's technical publication on algorithm; Page dedicated to developers (e.g., APIs); Whitepapers; 510K

Table A4. Definitions of Concepts		
Variable	Definition	Sources
Intelligent Algorithm	Capable of learning from big data inputs and changing its behaviors and outcomes dynamically	Benbya et al. (2021)
Algorithmic social injustice	An algorithm systematically and unfairly discriminates against certain individuals or groups of individuals in favor of others.	Friedman and Nissenbaum (1996)
Damage to user organization	<i>Harm to people</i> : Algorithm causes physical, digital, or psychological negative effect on customers, employees, or other stakeholders of the user organization that can result in mistreatment, loss of finances, disruption in daily life activities, or adverse to mental well-being.	Agrafiotis et al. (2018)
	<i>Financial loss</i> : Algorithmic social injustices cause loss of customers, regulatory fines, or compensations to stakeholders.	Baer (2019)
	<i>Reputational damage</i> : Algorithmic social injustices cause bad press, resentment, and pressure from the public and advocacy groups, affecting the user organization's standing in society.	
	<i>Lawsuit</i> : Algorithmic social injustices that translate into violations banned by law, invoking legal action against the user organization.	
Algorithm's ground truth status	The sum of all the data that has been collected, checked, and labeled in the context of a specific decision task. It is considered to be true for training an algorithm on how to make decisions on the task and calibrating the algorithm's decision criteria, rules, and boundaries.	C3.ai (2022); Kozyrkov (2021); Lexico (2021)
IT failure in algorithm's ecosystem	A breakdown or malfunction in any component in the algorithm's IT ecosystem that renders the IT ecosystem incapable of performing its intended tasks.	Triche and Walden (2018)
Algorithm's model failure	Algorithm's analytical model fails to deliver acceptable accuracy and reliability in its outputs due to missing, wrong, incorrect proxies, or unexpected interactions amongst variables, or inadequate or unrealistic assumptions and relationships.	Padmanabhan et al. (2022)
User organization's algorithm stakeholder management quality	<i>Stakeholder relations</i> : Actions taken to examine the nature of the algorithm's stakeholder relationships to better understand how value is created with and for stakeholders within the existing relationships.	Freeman et al. (2017)
	<i>Communicates with stakeholders</i> : Interaction with the algorithm's stakeholders aimed at understanding the role of the social and political surroundings to answer stakeholder concerns and co-create mutual understanding among the organization and its stakeholders.	
	<i>Learns with and from stakeholders</i> : Engages with stakeholders to learn about complex activities and create joint value by considering the various stakeholders' interests.	
	<i>Integrative stakeholder engagement</i> : Focuses on the power of stakeholder relationships and the interconnections among stakeholders.	
Algorithm's scope of human-like capabilities	<i>Vision, speech, emotion, cognitive, and sensory algorithmic capabilities</i> : The process through which algorithms inspect and analyze images, understand and analyze the language used by humans, recognize emotions in human text, sense surroundings, or understand the meaning of sensory inputs, explaining what they are doing, intend, or have done.	Lake et al. (2017)
developer organization's governance and controls	<i>Board-level oversight of algorithmic risk</i> : Board is responsible for understanding, quantifying, and overseeing algorithmic risks.	Clinton (2020); Silverman (2020)
	<i>Commitment to ethical, responsible algorithm development</i> : Creates algorithms with good intentions to empower employees and organization and fairly impact customers and society.	Eitel-Porter (2020); Leslie (2019)

	<i>Algorithmic accountability</i> : Committed to algorithm's legal and ethical obligations. Ethically implements algorithms and remedies any algorithm failures faced by stakeholders.	U.S. Congress (2022); Koene et al. (2019)
	<i>Algorithmic transparency</i> : Provides insight to stakeholders into the internal operations of the algorithm and its ecosystem by tracing the path of input, logic, model, or decision variables to explain a specific input-output outcome of the algorithm.	Asatiani et al. (2021); Hosanagar and Jair (2018)
	<i>Algorithmic development team's diversity</i> : A work unit with a distribution of personal attributes of race, age, and gender amongst interdependent members.	Moore et al. (2021)
	<i>Adoption of FACT principles</i> : An approach by data science teams to develop algorithms in addressing scientific challenges, using fairness, accuracy, confidentiality, and transparency.	RDS (2022); van der Aalst et al. (2017)
	<i>Governance and control mechanisms over the algorithmic development process</i> : The mechanisms in place to achieve accountability, transparency, and technical assurances by reducing undesirable algorithmic outcomes while preserving the algorithm's effectiveness.	Babel et al. (2019)
	<i>Openness to algorithm audits by clients</i> : Assessing algorithms, data, and design processes while preserving the intellectual property of artificial intelligence systems.	Guszcza et al. (2018)
User organization's governance and controls	<i>Ethical use of an algorithm</i> : A set of values, principles, and techniques that employ accepted standards to guide moral conduct in an organization's use of an algorithm.	Leslie (2019)
	<i>Accountability for this algorithm</i> : Commitment to the algorithm's legal and ethical obligations by demonstrating ethical implementation of algorithms to internal and external stakeholders and rectifying the algorithm's failure to perform as intended.	Koene et al. (2019)
	<i>Transparency about this algorithm</i> : Providing stakeholders insights into the algorithm's path of input, logic, model, or decision variables to explain a specific input-output outcome.	Asatiani et al. (2021); Koene et al. (2019)
Algorithm's privacy breach	Algorithm violates the interest an individual has in controlling or at least significantly influencing the handling of data about themselves by secondary use through the processing and dissemination of an individual's data or collection of data from an unaware individual.	Belanger et al. (2002); Clarke (1999)
Algorithm's cybersecurity breach	A malfunction in the algorithm or algorithm ecosystem due to malicious unauthorized access compromises the algorithm or its data's confidentiality, integrity, or availability. The breach may impact a single instance of the algorithm or the entire organization's infrastructure.	Samonas and Coss (2014)
Algorithm's stakeholder groups	Any group or individual that has an influence over the algorithm or is influenced by its objectives, in the form of power, legitimacy, or urgency relationship with the algorithm's developers.	(Mitchell et al., 1997)
Algorithm's decision support mode	The process algorithms use to support human decision-making based on the augmentation-automation continuum.	Teodorescu et al. (2021); Martin (2019)
Accuracy problem	The ratio of incorrect decisions to total number of decisions recommended by the algorithm	Lebovitz et al. (2021); Morse et al. (2022)

Table A5. Illustrative Examples of the Coding of Ground Truth, IT Failure, and Model Failure Variables

Coding	Algorithm	Evidence	Link
Well-established ground truth	TrueAllele—DNA probabilistic genotyping algorithm, 2016	The algorithm takes DNA samples and assesses the likelihood that a person's DNA matches that of a sample containing mixtures of genetic material to aid in criminal justice decision-making. <i>"DNA remains the gold standard for solving crimes, bolstered by academics, verified scientific studies, and experts."</i> TrueAllele algorithm uses DNA information as a well-established ground truth for training. "The data compiled by DNA analysts are fed into TrueAllele ... <i>it can learn from that data</i> , figure out the parameters ... there is no calibration needed with TrueAllele."	Kirchner (2016)
No established ground truth yet	Maricopa County Sheriff's Office (MSCO)—Predictive policing algorithm, 2019	The algorithm forecasts a region's criminal activity and allocates police resources. In Maricopa County, it was increasingly used in the context of local immigration enforcement. However, the data used to calibrate the algorithm's decision-making rules and boundaries were not adequately established for such a context. "An emerging trend of local police departments <i>engaging in immigration enforcement complicates</i> the oversight of police practices ... <i>the practices and policies of relevant police departments are ill-defined.</i> "	Richardson et al. (2019)
Multiple, conflicting ground truth	Amazon Marketplace—recommender algorithm, 2019	The algorithm takes input from a user and recommends books, including those claiming to cure autism or those with vaccination misinformation. The algorithm's decision-making rules and boundaries were trained with multiple and conflicting ground truths. "As a bookseller, we provide our customers with access to <i>various viewpoints</i> , including books that some customers may <i>find objectionable.</i> " The statement from an Amazon representative demonstrates the organization's intent to grant books customers want without regard to the safety or validity of the books. In an opposing viewpoint, the FDA responded to unproven autism cure books being sold on Amazon as follows: "it poses a significant risk to consumers who use this product for self-treatment."	Sarlin (2019)
IT failure	Fitbit—Fitness tracking algorithm, 2018	Fitbit's algorithm IT ecosystem has <i>optical sensors</i> to read vital signs from a person's skin and provide fitness information and accelerometers' electromagnetic sensors to pick up on motion. The optical sensors use green lights that must be reflected, but research suggests <i>IT malfunctions</i> when the light is absorbed by darker skin. As a result, the tracking algorithm gets inaccurate measurements that render it incapable of performing its intended tasks. Further, "the accelerometer measures only "normal" paced steps and cadence...without such motion, steps are not tracked."	Feehan et al. (2018)
Privacy breach	ByteDance—Text-to-speech algorithm, 2021	"ByteDance's text-to-speech (TTS) algorithm allows any text entered to automatically be converted to speech using a synthetic voice built from thousands of voice samples. It is then played over TikTok videos." The algorithm learns and imitates the voices of voice actors and actresses. ByteDance exploited the voice actress Beverly Standing's voice recordings to train the algorithm <i>without consent</i> , losing the ability to <i>control her voice's dissemination and association.</i>	Ferrarin (2021)
Cyber-security breach	Google—Image search algorithm, 2017	The algorithm returns image search results based on a text string input. Coordinated malicious actions by online anti-Semites manipulated the integrity dimension of the algorithm's cybersecurity. Malicious actors posted a specific term, such as "Jewish baby stroller," next to the image of ovens on wheels to resemble Jewish genocide. When users searched the term, the <i>algorithm was fooled into promoting</i> such mislabeled and hateful images.	Sales (2020)

Table A6. Illustrative Examples of the Coding of Stakeholder Management Quality Variables

Algorithm	Stakeholders	Creates value with (at least 50% of) Stakeholders	Understand social and political surroundings	Learns with and from the algorithm's stakeholders	Takes an integrative view of various stakeholders	Link
GoFundMe Search-Matching Algorithm, 2017— The algorithm takes text input from fundraisers to perform a search that creates lists of campaigns and funders and matches them with a fundraiser.	1. GoFundMe data science team 2. GoFundMe customer operations business unit 3. Fundraisers on GoFundMe 4. Fund suppliers on GoFundMe 5. Regulators who oversee financing activities 6. Media platforms	The algorithm was developed by GoFundMe's data science team and used by the Customer Operations business unit of GoFundMe. When a campaign becomes funded, the GoFundMe Customer operations unit gains value by a transaction fee. The search-matching algorithm brings together fundraisers, fund suppliers, and media platforms to create value for all three stakeholder groups. Successful campaigns provide financial relief to fundraisers, fund suppliers gain intrinsic value in helping others, and media platforms are provided with content to stimulate user engagement.	The search-matching algorithm considers fund suppliers' social and political surroundings to maximize good quality matches. For example, "In response to campaigns in natural disasters, [GoFundMe business unit] worked behind the scenes to make [fundraisers] efforts go viral, by constantly monitoring the platform for campaigns with a spike in shares or donations per minute. From there, they start pitching to local press [Media platforms] ... when a story starts to fade, [GoFundMe business unit] publishes a list of the top campaigns to increase exposure [to fund suppliers]."	GoFundMe works with fundraisers and fund suppliers to solve common problems. For example, it faced fraud in campaigns, it contacted the victimized families (e.g., Godwin family) to come up with the following learning to implement in the search-matching algorithm; (1) expects validation of fundraiser, (2) expects a single account, (3) makes the beneficiary transparent. GoFundMe spokesperson: "It's not uncommon for someone to create a GoFundMe after a news report. We have spoken with the Godwin family to guarantee the money will be deposited to a family member" "We empower our community [fundraisers and suppliers] to contact us if they see questionable activity."	For the search-matching algorithm to deliver joint value, GoFundMe needs to take into account the perspectives of all three sides of its platform: fundraiser, fund supplier, and media platforms. "We have a dedicated team [GoFundMe Customer Operation] that monitors fundraisers created to help individuals, families, and communities [side 1: fundraisers] affected by crises. To maximize the impact donors [side 2: fund suppliers] can have on the lives of others, we may share verified fundraisers with news organizations on our social channels [side 3: media platforms]."	Harris (2017) GoFundMe (2021) Willingham (2017)

Table A7. Illustrative Examples of the Coding of the Algorithm's Human-Like Capabilities

Algorithm	Vision	Speech	Emotion	Cognitive	Sensory	Link
Lee's Famous Recipe— Drive-thru AI, 2021	None	"Our technology is <i>conversational AI</i> . It's composed of <i>speech recognition</i> technology that can take the audio of your speech."	None	"It understands natural language and <i>meaning</i> of what customers want ... it <i>suggests items</i> and enters into the POS."	None	Duke (2021)
Tesla— Model Y autopilot driver monitoring, 2021	"A camera and radar help the car <i>understand the obstacles</i> ahead ... the radar <i>enables detection</i> of moving objects."	"Leverages a <i>natural language processor</i> that <i>interprets requests</i> and translates it to an action for the vehicle."	None	"The camera, radar, and sensors <i>work together</i> to <i>provide real-time feedback</i> ... sensors help <i>understand its environment</i> to steer itself safely."	"The Autopilot system is designed to <i>sense</i> if <i>your hands</i> are on the wheel and alerts you if you haven't <i>touched</i> the wheel."	Thompson (2016)
Sense.ly— Virtual nurse assistant, 2016	None	"Patients talk into the app, and the algorithm responds by <i>speaking with patients</i> ... <i>asks users</i> how they're doing with 5-minute check-ins."	" <i>Emotional analysis</i> helps it alert when patients experience <i>depression</i> or <i>anxiety</i> ...it responds to <i>emotions</i> with <i>empathy</i> ."	" <i>Retains relationship memory</i> and predicts a user's intent, simply by exploring the language they use."	None	Mack (2017) Kolodny (2017)
Sentri360— End-to-end workplace safety, 2021	"Uses <i>vision capabilities</i> for safety zone compliance of workers positioning."	None	None	"Sentri360 scans sensor inputs in real-time and <i>proactively alerts</i> workers to safety concerns."	"Scans multi-sensor <i>inputs</i> to proactively alert workers through a <i>wearable</i> personal device."	NED (2020)
Amazon— Halo Band, 2020	"It asks you to take <i>photos</i> of your body with its app to <i>estimate</i> your body fat percentage."	" <i>Listens to your conversations</i> so you can <i>understand how you sound</i> to others ... Halo gives you <i>voice directions</i> through the measurement."	"Halo Band can track whether you're <i>amused</i> or <i>delighted</i> , <i>caring</i> or <i>worried</i> understanding <i>emotion</i> is key to health, so it analyzes energy and <i>positivity</i> ."	"Halo Band gives you <i>insights</i> on how you're perceived ... e.g., <i>revealing</i> that a difficult work call leads to less positivity with a customer's family."	None	Dignan (2020)

Table A8. Illustrative Examples of the Coding of Developer and User Organizations' Governance and Control Variables

Variables	Organization	Coding	Algorithm	Source of evidence for coding*	Link
Developer organization's BoD oversight of algorithmic risk	Lemonade Insurance	[0]: None	Risk assessment algorithm, 2017	Searches on Lemonade Insurance yielded no evidence of BoD-level oversight of algorithmic risks in 2017. Keywords: Algorithm, risk, committee, artificial intelligence, board of directors, BoD, board of trustees, board of governors, board of regents, DEF 14A, advisory board	FACTIVA, Google Search
	Rocket mortgage	[1]: Symbolic	Underwriting automation algorithm, 2019	Rocketmortgage public website (2019): Two board members (Jonathan Mariner and Nancy Tellem) have significant technology risk management experience. But there is no board committee with algorithmic risk oversight.	Rocket Companies (2019)
	Google	[2]: Substantive	Vision cloud algorithm, 2020	Google proxy statement (2020): Evidence of a board committee responsible for AI risk management and oversight.	Alphabet (2020)
Developer organization's commitment to ethical, responsible algorithm development	Google	[1]: Symbolic	Advertising algorithm, 2017	Google annual report (2017): Statement of responsible algorithm development, but no substantive details: "We face challenges from low-quality and irrelevant content websites, including content farms. We are continually launching algorithmic changes focused on low-quality websites."	Alphabet (2017)
	Uber	[2]: Substantive	Dynamic pricing algorithm, 2020	Uber website (2020): Evidence of an AI research team and its publications on how to develop ethical AI.	Wiggers (2020)
Developer organization's algorithmic accountability	Proctorio	[1]: Symbolic	Exam monitoring algorithm, 2020	Proctorio website (2020): Proctorio claims to have algorithmic accountability for the security and privacy of test-taker data, but not for biased outcomes: "Keeping test-taker information and data secure is our top priority. This means making an effort to protect recording and personally identifiable information."	Proctorio (2020)
	IBM	[2]: Substantive	Cancer treatment recommender algorithm, 2019	News article (2019): Evidence that IBM committed to algorithmic accountability: "Intended to address the concerns about algorithm accountability, IBM announced the 'Drift Detection' capability to detect bias."	Black (2019)
Developer organization's algorithmic transparency	Scale AI	[1]: Symbolic	Image recognition algorithm, 2021	Scale AI website (2021): Some transparency about data labeling, but none about how the algorithm processes the data and generates outputs: "Image tasks submitted to the platform are first pre-labeled by our proprietary ML models, then manually annotated and reviewed by highly trained workers."	ScaleAI (2021)
	Facebook	[2]: Substantive	Advertisement algorithm, 2021	Facebook website (2021): Evidence of transparency about Facebook's algorithms: "We look forward to discussing more how our models work with outside experts and regulators—and preserving privacy. We launched a new online Transparency Center in May, publishing information about algorithms."	Presenti (2021)
Diversity of developer organization's algorithmic development teams	Clearview AI	[0]: None	Surveillance algorithm, 2020	Searches on Clearview AI yielded no evidence to suggest that the organization attempted to diversify its algorithm development teams in 2020. For example, all developers on Clearview AI's software team were male. Further, there were no statements by Clearview AI about attempting to diversify their team. Keywords: Team diversity, technology team, meet the team	Clearview (2020)
	Instructure	[1]: Symbolic	Cheating detection algorithm, 2021	Instructure website (2021): Evidence of Instructure taking symbolic actions to diversify algorithm development teams, but no evidence of an existing diverse team. SVP of Engineering: "I'd encourage people to get involved in the movement for diversity in engineering and apply on the career page."	Instructure (2021)
	Qoves	[2]: Substantive	Beauty algorithm, 2019	Qoves website (2019): In a team of 12, there were four females and three men of color. Further, diversity was actively sought: "We're always on the lookout for fresh faces (pun intended)."	Qoves (2020)

Developer organization's adoption of fact principles	Twitch	[1]: Symbolic	Chat filtering algorithm, 2021	Twitch transparency report (2021): The data science team publishes a report that displays transparency and accountability: "Transparency is an important measure of accountability that provides insight about how we are improving safety on Twitch."	Twitch (2021)
	Microsoft	[2]: Substantive	Cortana voice assistant algorithm, 2020	Microsoft website (2020): Microsoft has a website dedicated to F.A.C.T principles used by its development teams: "We aim to facilitate computational techniques that are innovative and responsible, prioritizing issues of fairness, accountability, transparency, and ethics as they relate to AI and ML."	Microsoft (2020)
Governance and controls of developer organization over its algorithmic development process	Snapchat	[1]: Symbolic	Advertising algorithm, 2019	Snapchat 10K (2019): Disclosure of concerns for general IT risk and controls, but not specific to algorithm risks and controls: "If our operations continue to grow, our information technology and internal controls may not adequately support our operations."	SEC (2019)
	Walmart	[2]: Substantive	Pricing algorithm, 2021	Walmart ESG report (2021) Evidence of corporate initiative for governing and controlling algorithm development process: "Walmart's Digital Citizenship team has developed frameworks to evaluate AI and machine learning models to mitigate bias in the development and implementation."	Walmart (2021)
Developer organization's openness to algorithm audits by clients	Tinder	[1]: Symbolic	Matching recommender algorithm, 2019	Tinder website (2019): Evidence of external audits focusing on cybersecurity and financial reporting, but no audits specific to algorithms: "We undergo yearly audits by independent firms to ensure our compliance with SOX and PCI-DSS."	Tinder (2019)
	Google	[2]: Substantive	Content moderation algorithm, 2020	News article (2020): Evidence of substantive focus on algorithm audits: "Google and the Partnership on AI have created a framework to help audit AI systems."	Johnson (2020)
User organization's ethical use of this algorithm	Adobe	[0]: None	Sensei and Photomerge Algorithm, 2018	Searches on Sensei and Photomerge yielded no evidence that Adobe used the algorithm ethically in 2018. Keywords: Adobe Sensei, ethics, ethical usage, fair, safe, legal	FACTIVA, Google Search
	Boston PD	[1]: Symbolic	Drone detection algorithm, 2021	City of Boston website (2021): Evidence of an ethical technology use policy, but no measures specific to the DroneShield Drone Detection Algorithm: "Employees can only use City technology for legal and ethical purposes."	City of Boston (2021)
	City of San Francisco	[2]: Substantive	Policing algorithm, 2019	News article (2019): Evidence of ethical use of policing algorithm: "We wanted to create something that can ensure that race is not going to play a role in our decision-making process."	Cutler (2019)
User organization's accountability for this algorithm	Teramind	[0]: None	Anomaly detection algorithm, 2019	Searches on anomaly-detecting algorithm yielded no evidence of a commitment to accountability for the algorithm in 2019. Keywords: anomaly-detection algorithm, accountability	FACTIVA, Google Search
	Uber	[1]: Symbolic	Background checks automation algorithm, 2019	Uber annual report (2019): Uber acknowledged algorithm concerns, suggesting accountability, but did not substantively address the concerns: "There have been allegations, including from regulators, legislators, taxicab owners, and consumers, that our background check process is insufficient or inadequate."	Uber (2019)
	YouTube	[2]: Substantive	Autocomplete algorithm, 2017	News article (2017): YouTube took full accountability for its algorithm endangering children: "This is an awful autocomplete result, and we really appreciate you making us aware. We've removed it and are working to remove similar results."	Ghosh (2017)
User organization's transparency about this algorithm	Bumble	[0]: None	Matching recommender algorithm, 2018	A search algorithm yielded no evidence of transparency in 2018. Keywords: Bumble, matching algorithm, recommender, transparency, transparent, data, model, logic, evidence	FACTIVA, Google Search
	Tesla	[1]: Symbolic	Autonomous driving algorithm, 2018	Tesla website (2018): General statements on algorithms, but no substantive transparency on its workings: "Autopilot requires driver supervision...full autonomy is dependent on reliability above human drivers and regulatory approval."	Tesla (2018)
	Argo AI	[2]: Substantive	Autonomous Driving Algorithm, 2021	Argo AI safety report (2021): Strong transparency evidence: "This report provides additional transparency...using ISO 21448 (SOTIF), we apply STPA and other safety approaches to identify design flaws in autonomous vehicle behavior."	ArgoAI (2021)

Note: *See Appendix Table A3 for details of the various organizational document sources used in coding.

Copyright of MIS Quarterly is the property of MIS Quarterly and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.